



3S COMMUNICATION PROTOCOL

Version 2.29

[Abstract](#)

Spectra Embedded
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INTRODUCTION

LEGENDS

1. XXh: any one-byte number in hexadecimal format.
2. XXd: any one-byte number in decimal format.
3. XXXXXXXXb: any one-byte number in binary format.
4. XYYh: any two-byte number (always in hexadecimal format).
5. AABCCDDh: any four-byte number (always in hexadecimal format).
6. CID: Controller ID.
7. UID: Unique Identification Number. (I.e. Card ID).
8. EXTID: Extended Card ID (Currently not in use, can be use to incorporate 7 byte UID)
9. SC: Site code of user.
10. RFU: Reserved for future use. Should be always zero.
11. FC: Function Code
12. LRC: Longitudinal Redundancy Check. Algorithm for LRC is stated as below...
13. LSB: Lower Significant Byte.
14. HSB: Higher Significant byte.
15. USB: upper significant byte.
16. MSB: Most significant byte.
17. R/W: Read or write flag.
18. TZ: Time Zone
19. APB: Anti Pass Back

PROTOCOL DEFINITION

Commands are sent to and are received from Device must follow this sequence.

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	YYh where $(YYd = (n+1) \% 100d)$
2	Total Length (HSB)	XXh where $(XXd = (n+1) / 100d)$
3	Command Or Response	01h : command (use this) 02h : response Any other value invalid
4	RFU	01h (fixed value)
5	Read Write Flag	00h : Read 01h : Write
6	FC (Function Code)	Predefined Function Number
7	Response Required?	00h: Required 01h: Not required
8	Payload Length (LSB)	YYh where $(YYd = (n-12) \% 100d)$
9	Payload Length (HSB)	XXh where $(XXd = (n-12) / 100d)$
10 to n-2	Payload	useful data in this frame
n - 1	LRC	LRC of 0 to n-2
n	Footer	Fixed BBh

Response to above commands should follow this sequence (**in case of no payload is there**)

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	0Ch (12d)
2	Total Length (HSB)	00h
3	Command Or Response	01h : command (use this) 02h : response Any other value invalid
4	RFU	01h (fixed value)
5	Read Write Flag	00h : Read 01h : Write
6	FC (Function Code)	Commanded Function Code
7	Response Code	XXh
8	Payload Length (LSB)	00h
9	Payload Length (HSB)	00h
10	LRC	
11	Footer	Fixed BBh

Response to above commands should follow this sequence (**with payload**)

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	YYh where $(YYd = (n+1) \% 100d)$
2	Total Length (HSB)	XXh where $(XXd = (n+1) / 100d)$
3	Command Or Response	01h : command (use this)

		02h : response Any other value invalid
4	RFU	01h (fixed value)
5	Read Write Flag	00h : Read 01h : Write
6	FC (Function Code)	Commanded Function Code
7	Response Code	XXh
8	Payload Length (LSB)	YYh (where YYd = length of Payload % 100d)
9	Payload Length (HSB)	XXh (where XXd = length of Payload / 100d)
9 to n - 2	Payload	Useful Data
n - 1	LRC	
n	Footer	Fixed BBh

EXAMPLES:

Get Status Command (FC: 01h) (read flag)

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	0Ch (12d)
2	Total Length (HSB)	00h
3	Command Or Response	01h : command (use this)
4	RFU	01h (fixed value)
5	Read Write Flag	00h : Read
6	FC (Function Code)	01h : get status command
7	Response Required?	00h (required)
8	Payload Length (LSB)	00h
9	Payload Length (HSB)	00h
10	LRC	
11	Footer	Fixed BBh

Get Status Command Response (response to read flag) (with payload)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	1Bh (27d)
2	Total Length (HSB)	00h
3	Command Or Response	02h: Response
4	RFU	01h
5	R/W flag	01h
6	FC	08h
7	Response	01h: Success
8	Payload length (LSB)	0Fh
9	Payload length (HSB)	00h
10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h

15	MAC ID 6	66h
16	IPv4 address segment 1	C0h (if ip is 192.100.100.90)
17	IPv4 address segment 2	64h
18	IPv4 address segment 3	64h
19	IPv4 address segment 4	5Ah
20	CID LSB	YYh (if CID is XYYh)
21	CID HSB	XXh
22	Year	00h to 63h
23	Month	01h to 0Ch
24	Date	01h to 1Fh
25	Day	01h to 7h
26	Hour	00h to 17h
27	Minute	00h to 3Bh
28	Second	00h to 3Bh
29	Communication Interface	1h: ETH 2h: WIFI 3h: GSM
30	RFU	0h
31	RFU	0h
32	Total Users (LSB)	DDh (four byte value AABBCDDh)
33	Total Users (HSB)	CCh
34	Total Users (USB)	BBh
35	Total Users (MSB)	AAh
36	Total Trans. (LSB)	DDh (four byte value AABBCDDh)
37	Total Trans. (HSB)	CCh
38	Total Trans. (USB)	BBh
39	Total Trans. (MSB)	AAh
40	Free Scan Finger LSB	
41	Free Scan Finger HSB	
42	Free Scan Finger USB	
43	Free Scan Finger MSB	
44	RFU	
45	RFU	
46	RFU	
47	RFU	
48 to 63	Firmware Version String Char 1 to 16	16 Char ASCII string
64 to 79	Protocol Version String Char 1 to 16	16 Char ASCII string
80	LRC	
81	Footer	Fixed BBh

Set Date Time command (02h) (with payload) (write flag)

Byte Index	Data	Value
0	Header	AAh
1	Total length (LSB)	13h (19d)
2	Total length (HSB)	00h
3	Command Or Response	01h (command)

4	RFU	01h
5	R/W flag	01h
6	FC	02h
7	Response Required?	00h (required)
8	Payload length (LSB)	07h
9	Payload length (HSB)	00h
10	Hour	00h to 17h
11	Minute	00h to 3Bh
12	Second	00h to 3Bh
13	Day	01h to 7h
14	Date	01h to 1Fh
15	Month	01h to 0Ch
16	Year	00h to 63h
17	LRC	
18	Footer	Fixed BBh

DATA HANDLING RULES

Data types having more than 8bits (1bytes) like word and long are identified and broken down as per follow in this document.

Multiple byte are organized as little endian convention. If data type of long having value of AABCCDDh. Then AA is MSB, BB is USB, CC is HSB and DD is LSB. More information on <https://en.wikipedia.org/wiki/Endianness>.

PREDEFINED MACROS

FP MODULE ERROR FLAGS

Error flags	Number
FP success	61h
Scan success	62h
Scan fail	63h
Finger not exist	69h
Finger not match	6Ah
Finger not good	6Bh
Command timeout	6Ch
Memory full	6Dh
Finger id already exists	6Eh
Finger Limit	72h
Continue	74h
Command not supported /parameter not matching	75h
Invalid id	76h
Matching timeout	7Ah
Busy	80h
Canceled	81h
Data error	82h
Data OK	83h
Finger already exists	86h
Duress finger	91h
Module locked	A1h
Id rejected	90h
Finger limit	94h
Fake finger	B0h

FP MODULE PARAMETER LIST

Parameter	Parameter Number	Possible Values
Finger Placement Time Out	62h	*3Ah: 10 seconds 3Bh ... 44h: 20 seconds
Template Size	64h	256d to *384d (in multiple of 32)
Security Level	66h	33h: 1/10000 34h: 1/100000 35h: 1/1000000 36h: 1/10000000 37h: 1/100000000 43h: 3/10000 44h: 3/100000 45h: 3/1000000 46h: 3/10000000 47h: 3/100000000

		*50h: Automatic Normal 51h: Automatic Secure 52h: Automatic More Secure
Number Of Enrolled Finger	73h (read only)	Will return number of free scan fingers
Available Fingers	74h (read only)	Will return free space
Maximum rotation	78h	31h: 15 degree *32h: 30 degree 33h: 45 degree 34h: 60 degree 35h: 75 degree 36h: 90 degree
Sensor Sensitivity	80h	30h: least sensitive .. *35: sensitivity 5 37h: most sensitive
Image quality	81h	*35h: moderate V2 36h: strong V2 37h: strongest V2
Free Scan	84h	0h: Disabled 10d to 60d: Enabled with timeout in seconds FFFFFFFFh : infinite
Matching Timeout	88h	*33h: 3 seconds ... 44h: 20 seconds
Lightning condition	90h	*30h: out door 31h: in door
Fast mode	93h	30h: normal 31h: fast 1 32h: fast 2 33h: fast 3 34h: fast 4 35h: fast 5 *36h: auto
Template Type	96h	30h: Suprema *31h: ISO

PHYSICAL ADDRESS FOR INPUTS

Available Inputs Per Controllers, other values are RFU / invalid.

BIOSTAMP_3S / BIOSCRIBE_3S :: Reader Port 1, 2. Switch 1. Door Sensor 1. Temper 1,2. GPI 1,2.

TWINXS_3S :: Reader Port 1,2,3,4. Switch 1,2. Door Sensor 1,2. Temper 1,2,3,4. GPI 1,2,3,4,5,6,7,8.

Input	Address
GPI 1	1d
GPI 2	2d
GPI 3	3d
GPI 4	4d
GPI 5	5d
GPI 6	6d

GPI 7	7d
GPI 8	8d
GPI 9	9d
GPI 10	10d
GPI 11	11d
GPI 12	12d
Reader Port 1	17d
Reader Port 2	18d
Reader Port 3	19d
Reader Port 4	20d
Reader Port 5	21d
Reader Port 6	22d
Reader Port 7	23d
Reader Port 8	24d
Temper 1	25d
Temper 2	26d
Temper 3	27d
Temper 4	28d
Temper 5	29d
Temper 6	30d
Temper 7	31d
Temper 8	32d
Switch 1	33d
Switch 2	34d
Switch 3	35d
Switch 4	36d
Switch 5	37d
Switch 6	38d
Switch 7	39d
Switch 8	40d
Door Sensor 1	41d
Door Sensor 2	42d
Door Sensor 3	43d
Door Sensor 4	44d
Door Sensor 5	45d
Door Sensor 6	46d
Door Sensor 7	47d
Door Sensor 8	48d

PHYSICAL ADDRESS FOR OUTPUTS

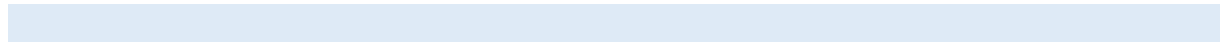
Available Outputs Per controllers. Other values are RFU / invalid

BIOSTAMP_3S / BISCRIBE_3S :: Lock 1. GPO 1.

TWINXS_3S :: Relay 1,2,3,4,5,6,7,8.

Output	Address

Relay 1 / Lock 1	51d
Relay 2 / GPO 1	52d
Relay 3	53d
Relay 4	54d
Relay 5	55d
Relay 6	56d
Relay 7	57d
Relay 8	58d



SOFTWARE EVENT CODES

Event	Address
Access Allowed	1d
Exceptional Allow	2d
Soft Local APB	3d
Soft Global APB	4d
Soft Local Timed APB	5d
Soft Global Timed APB	6d
Soft Conditional APB	7d
Soft Concentric APB	8d
Soft Matrix APB	9d
Soft Local Access Limit	10d
Soft Global Access Limit	11d
Soft Local User Absent	12d
Soft Global User Absent	13d
Soft Custodian Not Found	14d
Soft Local Occupancy	15d
Soft Global Occupancy	16d
Soft Local Sentry Check	17d
Soft Global Sentry Check	18d
	19d
	20d
	21d
	22d
	23d
	24d
	25d
	26d
	27d
	28d
	29d
	30d
	31d
	32d
Unknown Card format	33d
Invalid Door Config	34d
User Not Found	35d
Invalid Site Code	36d
Validity Not Started	37d
Validity Expired	38d
Not On This Door	39d
Out Of Time Zone	40d
Due To Holiday	41d
Due To Week Off	42d
Wait For Escort	43d
Escort Fail	44d
User Blocked	45d
Finger Scan Failed	46d
Finger not Found	47d
Finger not matched	48d
Finger not good	49d
Finger timeout	50d
Fp command error	51d

Finger before card	52d
Finger not allowed	53d
Invalid pin	54d
Pin entry timeout	55d
Forcefully closed	56d
Switch pressed	57d
DotI event	58d
ForceFull event	59d
Server deny	60d
Server fail	61d
Local apb fail	62d
Global apb fail	63d
Local timed apb fail	64d
Global timed apb fail	65d
Conditional apb fail	66d
Concentric apb fail	67d
Matrix apb fail	68d
Local access limit	69d
Global access limit	70d
Local user absent	71d
Global user absent	72d
No custodian found	73d
Local occupancy limit	74d
Global occupancy limit	75d
Local sentry check fail	76d
Global sentry check fail	77d
Further authentication needed	78d
Door interlocked	79d
Sentry can not pass	80d
Fake finger	81d
Daily Item Limit Exceeded	101d
Weekly Item Limit Exceeded	102d
Monthly Item Limit Exceeded	103d
TimeZone wise Item Limit Exceeded	104d
Balance Limit Exceeded	105d

RESPONSE CODE LIST

Output	Address
Success /OK	1d
Invalid Frame Length	2d
LRC Check Fail	3d
Invalid Data Arguments	4d
Invalid Payload Length	5d
Invalid Response	6d
Invalid Header	7d
Invalid Footer	8d
Too Large Frame	9d
Long Timeout	10d
Unknown Command	11d
No CID Set	12d

Write Error	13d
Read Error	14d
User Not Found	15d
Unknown Error	16d
Invalid Range	17d
Require Read only Operation	18d
Require Write Only Operation	19d
Invalid Sequence	20d
CRC Match Fail	21d
Device Busy	22d
Command Timeout	23d
Data Full	24d
Not Ready	25d

FUNCTION CODE (WITH PAYLOADS)

GET STATUS (01H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h / 02h As Identifier string device will send this frame as command.
4	RFU	01h
5	R/W flag	00h
6	FC	01h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	IPv4 address segment 1	C0h (if ip is 192.100.100.90)
17	IPv4 address segment 2	64h
18	IPv4 address segment 3	64h
19	IPv4 address segment 4	5Ah
20	CID LSB	YYh (if CID is XXYYh)
21	CID HSB	XXh
22	Year	00h to 63h
23	Month	01h to 0Ch
24	Date	01h to 1Fh
25	Day	01h to 7h
26	Hour	00h to 17h
27	Minute	00h to 3Bh
28	Second	00h to 3Bh
29	Communication Interface	1h: ETH 2h: WIFI 3h: GSM
30	RFU	0h
31	RFU	0h
32	Total Users (LSB)	DDh (four byte value AABCCDDh)
33	Total Users (HSB)	CCh
34	Total Users (USB)	BBh
35	Total Users (MSB)	AAh
36	Total Trans. (LSB)	DDh (four byte value AABCCDDh)
37	Total Trans. (HSB)	CCh
38	Total Trans. (USB)	BBh
39	Total Trans. (MSB)	AAh
40	Free Scan Finger LSB	
41	Free Scan Finger HSB	
42	Free Scan Finger USB	

43	Free Scan Finger MSB	
44	RFU	
45	RFU	
46	RFU	
47	RFU	
48 to 63	Firmware Version String Char 1 to 16	16 Char ASCII string
64 to 79	Protocol Version String Char 1 to 16	16 Char ASCII string
80	LRC	
81	Footer	Fixed BBh

SET DATE & TIME (02H)

Byte Index	Data	Value
0	Header	AAh
1	Total length (LSB)	YYh
2	Total length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	02h
7	Response Required? /Response	01h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Hour	00h to 17h
11	Minute	00h to 3Bh
12	Second	00h to 3Bh
13	Day	01h to 7h
14	Date	01h to 1Fh
15	Month	01h to 0Ch
16	Year	00h to 63h
17	LRC	
18	Footer	Fixed BBh

ADD USER (TIME ZONE BASED ACCESS) (03H)

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	03h
7	Response Required? /Response	00h / XXh

8	Payload Length (LSB)	YYh
9	Payload Length(HSB)	XXh
10	User Checking Bypass	0h: No Search 1h: Search
11	UID LSB	78h(If UID is 0x12345678)
12	UID HSB	56h
13	UID USB	34h
14	UID MSB	12h
15	EXTD ID LSB (for 64 bit CardID only)	78h(If UID is 0x12345678)
16	EXTD ID HSB	56h
17	EXTD ID USB	34h
18	EXTD ID MSB	12h
19	Time Zone Wise / Exclusive	1h: Tz wise (fixed)
20	User Type	0d to 32d (0h to 20h)
21	Week Off group number LSB	Week Off Group ID
22	Week Off group number HSB	
23	Holiday Access group LSB	Holiday Group ID
24	Holiday Access group HSB	
25	Start Year	Validity start time
26	Start Month	
27	Start Date	
28	Start Hour	
29	Start Minute	
30	End Year	Expiry time
31	End Month	
32	End Date	
33	End Hour	
34	End Minute	
35	Pin LSB	4Eh (if pin is 12345678d == BC614Eh)
36	Pin HSB	61h
37	Pin USB	BCh
38	Pin MSB	00h
39 to 70	User Name (32 Bytes)	"Name of User" ASCII string null terminated.
71 to 86	Employee ID (16 Bytes)	ASCII string null terminated.
87	Time Zone Group ID 1 LSB	Group No of Time Zone.
88	Time Zone Group ID 1 HSB	If any time zones not used, Fill 0h.
89	Time Zone Group ID 2 LSB	Max 8 time zones.
90	Time Zone Group ID 2 HSB	
91	Time Zone Group ID 3 LSB	Canteen Management System :
92	Time Zone Group ID 3 HSB	
93	Time Zone Group ID 4 LSB	Byte 87 : Item Group Number (1d to 100d)
94	Time Zone Group ID 4 HSB	BYTE 102 : Prepaid (1) / Postpaid (0)
95	Time Zone Group ID 5 LSB	
96	Time Zone Group ID 5 HSB	
97	Time Zone Group ID 6 LSB	
98	Time Zone Group ID 6 HSB	
99	Time Zone Group ID 7 LSB	
100	Time Zone Group ID 7 HSB	
101	Time Zone Group ID 8 LSB	
102	Time Zone Group ID 8 HSB	
103	Escort User ID LSB	
104	Escort User ID HSB	

105	Escort User ID USB	
106	Escort User ID MSB	
107	Escort User EXTID ID LSB (for 64 bit CardID only)	
108	Escort User EXTID ID HSB	
109	Escort User EXTID ID USB	
110	Escort User EXTID ID MSB	
111	LRC	LRC
112	Footer	Fixed BBh

ADD USER (EXCLUSIVE GATE WISE ACCESS) (04H)

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	04h
7	Response Required? /Response	00h / XXh
8	Payload Length (LSB)	YYh
9	Payload Length(HSB)	XXh
10	User Checking Bypass	0h: No Search 1h: Search
11	UID LSB	78h(If UID is 0x12345678)
12	UID HSB	56h
13	UID USB	34h
14	UID MSB	12h
15	EXTD ID LSB (for 64 bit CardID only)	78h(If UID is 0x12345678)
16	EXTD ID HSB	56h
17	EXTD ID USB	34h
18	EXTD ID MSB	12h
19	Time Zone Wise / Exclusive	2h: Exclusive Gate wise (fixed)
20	User Type	0d to 32d (1h to 20h)
21	Week Off group number LSB	Week Off Group ID
22	Week Off group number HSB	
23	Holiday Access group LSB	Holiday Group ID
24	Holiday Access group HSB	
25	Start Year	Validity start time
26	Start Month	
27	Start Date	
28	Start Hour	
29	Start Minute	
30	End Year	Expiry time
31	End Month	
32	End Date	
33	End Hour	
34	End Minute	
35	Pin LSB	4Eh (if pin is 12345678d == BC614Eh)

36	Pin HSB	61h
37	Pin USB	BCh
38	Pin MSB	00h
39 to 70	User Name (32 Bytes)	"Name of User" ASCII string null terminated.
71 to 86	Employee ID (16 Bytes)	ASCII string null terminated.
87	Door 1 Start Time Hour	0d to 23d
88	Door 1 Start Time Minute	0d to 59d
89	Door 1 End Time Hour	0d to 23d
90	Door 1 End Time Minute	0d to 59d
91	Door 2 Start Time Hour	
92	Door 2 Start Time Minute	
93	Door 2 End Time Hour	
94	Door 2 End Time Minute	
95	Door 3 Start Time Hour	
96	Door 3 Start Time Minute	
97	Door 3 End Time Hour	
98	Door 3 End Time Minute	
99	Door 4 Start Time Hour	
100	Door 4 Start Time Minute	
101	Door 4 End Time Hour	
102	Door 4 End Time Minute	
103	Escort User UID LSB	
104	Escort User UID HSB	
105	Escort User UID USB	
106	Escort User UID MSB	
107	Escort User EXTID ID LSB (for 64 bit CardID only)	
108	Escort User EXTID ID HSB	
109	Escort User EXTID ID USB	
110	Escort User EXTID ID MSB	
111	LRC	LRC
112	Footer	Fixed BBh

UPLOAD FILE HEADER (05H)

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	05h
7	Response Required? /Response	00h / XXh
8	Payload Length (LSB)	YYh
9	Payload Length(HSB)	XXh
10	Mode	0h: edit file 1h: create file 2h: delete file

11	RFU	
12	RFU	
13	RFU	
14	RFU	
15	File Size in Bytes LSB	E8h (If file size is 1000d = 3E8h)
16	File Size in Bytes HSB	03h
17	File Size in Bytes USB	00h
18	File Size in Bytes MSB	00h
19 to N - 2	File name including folder path	Null terminated Ascii string like "USER/aabbccdd.bmp\0" "AUDIO/allow.wav\0" "AUDIO/deny.wav\0" "complogo.bmp\0" "SS/SS01.jpg\0"
N - 1	LRC	LRC
N	Footer	Fixed BBh

UPLOAD FILE DATA (06H)

File must be created using 05h command first.

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	06h
7	Response Required? /Response	00h / XXh
8	Payload Length (LSB)	YYh (max 1024 for this command)
9	Payload Length(HSB)	XXh
10	Sequence Number LSB	DDh (if seq. No. = AABBCDDh)
11	Sequence Number HSB	CCh
12	Sequence Number USB	BBh
13	Sequence Number MSB	AAh
15 to N - 2	File Data	Binary Data max 1024 bytes
N - 1	LRC	LRC
N	Footer	Fixed BBh

DELETE PARTICULAR USER (07H)

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh

3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	07h
7	Response Required? /Response	00h / XXh
8	Payload Length (LSB)	YYh
9	Payload Length(HSB)	XXh
10	UID LSB	DDh (if card id is AABBCDDh)
11	UID HSB	CCh
12	UID USB	BBh
13	UID MSB	AAh
14	EXTD ID LSB (for 64 bit CardID only)	DDh (if card id is AABBCDDh)
15	EXTD ID HSB	CCh
16	EXTD ID USB	BBh
17	EXTD ID MSB	AAh
18	LRC	
19	Footer	Fixed BBh

DELETE ALL USERS (I.E. DB DELETE ALL) (08H)

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	08h
7	Response Required? /Response	00h / XXh
8	Payload Length (LSB)	00h
9	Payload Length(HSB)	00h
10	LRC	
11	Footer	Fixed BBh

ADD USER EXCEPTION (09H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	09h

7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Search or No Search	0h: do not search add new 1h: search for existing user
11	UID LSB	
12	UID HSB	
13	UID USB	
14	UID MSB	
15	Site Code LSB	
16	Site Code HSB	
17	Site Code USB	
18	Site Code MSB	
19	Door 1 Block Access / Forcefully Allow	0d: nothing 1d: block 2d: Force Allow
20	Door 1 Exception Start Year	0d to 99d
21	Door 1 Exception Start Month	1d to 12d
22	Door 1 Exception Start Date	1d to 31d
23	Door 1 Exception Start Hour	0d to 23d
24	Door 1 Exception Start Minute	0d to 59d
25	Door 1 Exception End Year	0d to 99d
26	Door 1 Exception End Month	1d to 12d
27	Door 1 Exception End Date	1d to 31d
28	Door 1 Exception End Hour	0d to 23d
29	Door 1 Exception End Minute	0d to 59d
30	Door 2 Block Access / Forcefully Allow	0d: nothing 1d: block 2d: Force Allow
31	Door 2 Exception Start Year	0d to 99d
32	Door 2 Exception Start Month	1d to 12d
33	Door 2 Exception Start Date	1d to 31d
34	Door 2 Exception Start Hour	0d to 23d
35	Door 2 Exception Start Minute	0d to 59d
36	Door 2 Exception End Year	0d to 99d
37	Door 2 Exception End Month	1d to 12d
38	Door 2 Exception End Date	1d to 31d
39	Door 2 Exception End Hour	0d to 23d
40	Door 2 Exception End Minute	0d to 59d
41	Door 3 Block Access / Forcefully Allow	0d: nothing 1d: block 2d: Force Allow
42	Door 3 Exception Start Year	0d to 99d
43	Door 3 Exception Start Month	1d to 12d
44	Door 3 Exception Start Date	1d to 31d
45	Door 3 Exception Start Hour	0d to 23d
46	Door 3 Exception Start Minute	0d to 59d
47	Door 3 Exception End Year	0d to 99d
48	Door 3 Exception End Month	1d to 12d
49	Door 3 Exception End Date	1d to 31d
50	Door 3 Exception End Hour	0d to 23d
51	Door 3 Exception End Minute	0d to 59d
52	Door 4 Block Access / Forcefully Allow	0d: nothing 1d: block

		2d: Force Allow
53	Door 4 Exception Start Year	0d to 99d
54	Door 4 Exception Start Month	1d to 12d
55	Door 4 Exception Start Date	1d to 31d
56	Door 4 Exception Start Hour	0d to 23d
57	Door 4 Exception Start Minute	0d to 59d
58	Door 4 Exception End Year	0d to 99d
59	Door 4 Exception End Month	1d to 12d
60	Door 4 Exception End Date	1d to 31d
61	Door 4 Exception End Hour	0d to 23d
	Door 4 Exception End Minute	0d to 59d
N - 1	LRC	
N	Footer	Fixed BBh

DELETE USER EXCEPTION (0AH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	0Ah
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	08h
9	Payload length (HSB)	00h
10	UID LSB	
11	UID HSB	
12	UID USB	
13	UID MSB	
14	Site Code LSB	
15	Site Code HSB	
16	Site Code USB	
17	Site Code MSB	
18	LRC	
19	Footer	Fixed BBh

SCAN LIVE FINGER (0BH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h

5	R/W flag	00h
6	FC	0Bh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	RFU	0h
11	Returned Template Size HSB	01h (if 384d = 180h)
12	Returned Template Size LSB	80h
13	Scanned Quality Score	Max 100d (in percentage)
14 to N-2	First Byte of Template Data	
...	...	
N - 2	Last Byte of Template Data	Max 512d bytes
N - 1	LRC	
N	Footer	Fixed BBh

SET FINGER TEMPLATE (0CH)

Finger Template Length may be variable. And depends upon finger size setting in fppara.

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	0Ch
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	No Of Templates in this command	1h or 2h
11	UID LSB	
12	UID HSB	
13	UID USB	
14	UID MSB	
15	EXTD ID LSB (for 64 bit CardID only)	
16	EXTD ID HSB	
17	EXTD ID USB	
18	EXTD ID MSB	
19	Template 1 Index	0h to 9h
20	Template 1 Scope / Template 2 index in Read command	0h: local 1h: free scan / 0h to 9h
21	Template 1 size LSB	80h (if 384d = 180h)
22	Template 1 size HSB	01h
23.. to M	Template 1 Data byte 0	
	...	

M	Template 1 Data byte last	
M+1	Template 2 Index	
M+2	Template 2 Scope	
M+3	Template 2 size LSB	
M+4	Template 2 size HSB	
M+5 to N - 2	Template 2 Data byte 0	
...	...	
N - 2	Template 2 Data byte last	
N - 1	LRC	
N	Footer	Fixed BBh

DELETE FINGER TEMPLATE (0DH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	0Dh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Scope	0h: local 1h: free scan
11	Template Index (RFU)	0h: fixed
12	UID LSB	
13	UID HSB	
14	UID USB	
15	UID MSB	
16	EXTD ID LSB (for 64 bit CardID only)	
17	EXTD ID HSB	
18	EXTD ID USB	
19	EXTD ID MSB	
20	LRC	
21	Footer	Fixed BBh

DELETE ALL FINGER (0EH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h

6	FC	0Eh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	01h
9	Payload length (HSB)	00h
10	Scope	0h: local 1h: free scan + local
11	LRC	
12	Footer	Fixed BBh

READ / WRITE FP MODULE PARAMETERS (0FH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h / 01h
6	FC	0Fh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Parameter No.	See FP para list
11	Value LSB	
12	Value HSB	
13	Value USB	
14	Value MSB	
15	LRC	
16	Footer	Fixed BBh

SYNC TIME ZONES (10H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	10h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh

10	TZ 1 Start Hour (fixed read only)	0d
11	TZ 1 Start Minute (fixed read only)	0d
12	TZ 1 End Hour (fixed read only)	23d
13	TZ 1 End Minute (fixed read only)	60d (60d value only valid for this byte)
14 to ...	...	0d to 59d
	...	0d to 59d
	...	0d to 23d
	...	0d to 59d
N - 5	TZ 127 Start Hour	0d to 59d
N - 4	TZ 127 Start Minute	0d to 59d
N - 3	TZ 127 End Hour	0d to 23d
N - 2	TZ 127 End Minute	0d to 59d
N - 1	LRC	
N	Footer	Fixed BBh

ADD TIME ZONE ACCESS GROUP (11H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	11h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
	No Of Time Zone Groups In This Frame	1d to 50d
10	Time Zone Access Group Number LSB	YYh (XXYYh = 2d to 1024d) (1d is read only)
11	Time Zone Access Group Number HSB	XXh
12	1st Time Zone for Gate 1	1d to 127d
13	2nd Time Zone for Gate 1	1d to 127d
14	3rd Time Zone for Gate 1	1d to 127d
15	4th Time Zone for Gate 1	1d to 127d
16	Allow / Deny for Gate 1	00h : allow 01h : deny
17	1st Time Zone for Gate 2	
18	2nd Time Zone for Gate 2	
19	3rd Time Zone for Gate 2	
20	4th Time Zone for Gate 2	
21	Allow / Deny for Gate 2	
22	1st Time Zone for Gate 3	
23	2nd Time Zone for Gate 3	
24	3rd Time Zone for Gate 3	
25	4th Time Zone for Gate 3	
26	Allow / Deny for Gate 3	

27	1st Time Zone for Gate 4	
28	2nd Time Zone for Gate 4	
29	3rd Time Zone for Gate 4	
30	4th Time Zone for Gate 4	
31	Allow / Deny for Gate 4	
32	LRC	
33	Footer	Fixed BBh

DELETE ALL TIME ZONE GROUPS (12H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	12h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	00h
9	Payload length (HSB)	00h
10	LRC	
11	Footer	Fixed BBh

SYNC WEEK OFF GROUPS (13H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	13h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Group 1 Week 1 Days	0b0XXXXXXX
11	Group 1 Week 2 Days	Where each X represents days 0b0SFTWTMS 1b: day on 0b: day off
12	Group 1 Week 3 Days	
13	Group 1 Week 4 Days	
14	Group 1 Week 5 Days	
15	...	
16	...	
17	...	

18	...	
19 to 164	...	
165	Group 32 Week 1 Days	
166	Group 32 Week 2 Days	
167	Group 32 Week 3 Days	
168	Group 32 Week 4 Days	
169	Group 32 Week 5 Days	
170	LRC	
171	Footer	Fixed BBh

SET HOLIDAY GROUP (14H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	14h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Holiday Group No.	1h to 32h
11	Date 1	1d to 31d
12	Month 1	1d to 12d
13	Year 1	00d to 99d
14	Date 2	
15	Month 2	
16	Year 2	
17 to ...	...	
N - 5	...	
N - 4	Date 128	
N - 3	Month 128	
N - 2	Year 128	
N - 1	LRC	
N	Footer	Fixed BBh

DELETE ALL HOLIDAY GROUPS (15H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h

5	R/W flag	01h
6	FC	15h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	00h
9	Payload length (HSB)	00h
10	LRC	
11	Footer	Fixed BBh

CONTROLLER SETUP (16H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	16h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	CID LSB	E8h (If CID is 1000d (3E8h))
11	CID HSB	03h
12	Device Mode	0h: Controller mode 1h: Wiegand Reader Mode
13	Reader Mode	0h: Default Mode 1h: In Mode 2h: Out Mode 3h: Toggle Mode 4h: Null Mode
14	RFU	0h
15	RFU	
16	RFU	0h
17	RFU	
18	RFU	0h
19	RFU	0h
20	RFU	0h
21	Transaction Frame Type	00h: Old Type Transaction Frame 01h: New Type Transaction Frame
22	LRC	
23	Footer	Fixed BBh

READER SETUP (17H)

Byte Index	Data	Value
0	Header	AAh

1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	17h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	CID LSB	E8h (If CID is 1000d (3E8h))
11	CID HSB	03h
12	Device Mode	1h: Wiegand Reader Mode
13	RFU	0h
14	Wigand Pulse Period LSB	50d to 1000d (micro seconds)
15	Wigand Pulse Period HSB	
16	Wigand Pulse Interval LSB	1000d to 10000d (micro seconds)
17	Wigand Pulse Interval HSB	
18	Authentication Type in Wiegand Reader Mode	01h: Card Only 02h: Card + Pin 03h: Card / Pin 04h: Card + Finger 05h: Card + Pin + Finger 06h: Only Finger 07h: Card / Finger
19	Transaction Push Enable?	0h: disable 1h: enable
20	RFU	0h
21	Transaction Frame Type	00h: Old Type Transaction Frame 01h: New Type Transaction Frame
22	LRC	
23	Footer	Fixed BBh

DOOR SETUP (18H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	18h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Door 1 Enable / Disable?	0h : disable 1h : enable
11	Door 1 Entry Reader Number	1h to 8h

12	Door 1 Exit Reader Number	
13	Door 1 Entry Mode	01h: Card Only 02h: Card + Pin 03h: Card / Pin 04h: Card + Finger 05h: Card + Pin + Finger 06h: Only Finger 07h: Card / Finger
14	Door 1 Exit Mode	
15	Door 1 Entry Relay	<u>See Output List</u>
16	Door 1 Exit Relay	
17	Door 1 Exit Verification for TZ	0h: disable *1h: enable
18	Door 1 Entry Open Time LSB	YYh (5d to 9999d) (seconds)
19	Door 1 Entry Open Time HSB	XXh
20	Door 1 Exit Open Time LSB	
21	Door 1 Exit Open Time HSB	
22	Door 1 Sensor Time Limit LSB	YYh (5d to 9999d) (seconds)
23	Door 1 Sensor Time Limit HSB	XXh (0d to disable feature)
24	Door 1 Monitor Enable/Disable?	0h : disable 1h : enable
25	RFU	0h
26	Door 2 Enable / Disable?	0h : disable 1h : enable
27	Door 2 Entry Reader Number	
28	Door 2 Exit Reader Number	
29	Door 2 Entry Mode	
30	Door 2 Exit Mode	
31	Door 2 Entry Relay	
32	Door 2 Exit Relay	
33	Door 2 Exit Verification for TZ	
34	Door 2 Entry Open Time LSB	
35	Door 2 Entry Open Time HSB	
36	Door 2 Exit Open Time LSB	
37	Door 2 Exit Open Time HSB	
38	Door 2 Sensor Time Limit LSB	
39	Door 2 Sensor Time Limit HSB	
40	Door 2 Monitor Enable/Disable?	0h : disable 1h : enable
41	RFU	0h
42	Door 3 Enable / Disable?	
43	Door 3 Entry Reader Number	
44	Door 3 Exit Reader Number	
45	Door 3 Entry Mode	
46	Door 3 Exit Mode	
47	Door 3 Entry Relay	
48	Door 3 Exit Relay	
49	Door 3 Exit Verification for TZ	
50	Door 3 Entry Open Time LSB	
51	Door 3 Entry Open Time HSB	
52	Door 3 Exit Open Time LSB	
53	Door 3 Exit Open Time HSB	
54	Door 3 Sensor Time Limit LSB	
55	Door 3 Sensor Time Limit HSB	
56	Door 3 Monitor Enable/Disable?	0h : disable

		1h : enable
57	RFU	0h
58	Door 4 Enable / Disable?	
59	Door 4 Entry Reader Number	
60	Door 4 Exit Reader Number	
61	Door 4 Entry Mode	
62	Door 4 Exit Mode	
63	Door 4 Entry Relay	
64	Door 4 Exit Relay	
65	Door 4 Exit Verification for TZ	
66	Door 4 Entry Open Time LSB	
67	Door 4 Entry Open Time HSB	
68	Door 4 Exit Open Time LSB	
69	Door 4 Exit Open Time HSB	
70	Door 4 Sensor Time Limit LSB	
71	Door 4 Sensor Time Limit HSB	
72	Door 4 Monitor Enable/Disable?	0h : disable 1h : enable
73		
74	LRC	
75	Footer	Fixed BBh

DOOR MODE EXCEPTIONS (19H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	19h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Door 1 Enabled Days	0b0XXXXXXX (0b0SMTWTFS)
11	Door 1 Mode Sunday	00h: Default Gate wise
12	Door 1 Mode Monday	01h: Card Only
13	Door 1 Mode Tuesday	02h: Card + Pin
14	Door 1 Mode Wednesday	03h: Card / Pin
15	Door 1 Mode Thursday	04h: Card + Finger
16	Door 1 Mode Friday	05h: Card + Pin + Finger
17	Door 1 Mode Saturday	06h: Only Finger 07h: Card / Finger
18	Door 1 Sunday Start 1 Hour	
19	Door 1 Sunday Start 1 Minute	
20	Door 1 Sunday End 1 Hour	
21	Door 1 Sunday End 1 Minute	
22	Door 1 Sunday Start 2 Hour	
23	Door 1 Sunday Start 2 Minute	
24	Door 1 Sunday End 2 Hour	

25	Door 1 Sunday End 2 Minute	
26	Door 1 Monday Start 1 Hour	
27	Door 1 Monday Start 1 Minute	
28	Door 1 Monday End 1 Hour	
29	Door 1 Monday End 1 Minute	
30	Door 1 Monday Start 2 Hour	
31	Door 1 Monday Start 2 Minute	
32	Door 1 Monday End 2 Hour	
33	Door 1 Monday End 2 Minute	
34	Door 1 Tuesday Start 1 Hour	
35	Door 1 Tuesday Start 1 Minute	
36	Door 1 Tuesday End 1 Hour	
37	Door 1 Tuesday End 1 Minute	
38	Door 1 Tuesday Start 2 Hour	
39	Door 1 Tuesday Start 2 Minute	
40	Door 1 Tuesday End 2 Hour	
41	Door 1 Tuesday End 2 Minute	
42	Door 1 Wednesday Start 1 Hour	
43	Door 1 Wednesday Start 1 Minute	
44	Door 1 Wednesday End 1 Hour	
45	Door 1 Wednesday End 1 Minute	
46	Door 1 Wednesday Start 2 Hour	
47	Door 1 Wednesday Start 2 Minute	
48	Door 1 Wednesday End 2 Hour	
49	Door 1 Wednesday End 2 Minute	
50	Door 1 Thursday Start 1 Hour	
51	Door 1 Thursday Start 1 Minute	
52	Door 1 Thursday End 1 Hour	
53	Door 1 Thursday End 1 Minute	
54	Door 1 Thursday Start 2 Hour	
55	Door 1 Thursday Start 2 Minute	
56	Door 1 Thursday End 2 Hour	
57	Door 1 Thursday End 2 Minute	
58	Door 1 Friday Start 1 Hour	
59	Door 1 Friday Start 1 Minute	
60	Door 1 Friday End 1 Hour	
61	Door 1 Friday End 1 Minute	
62	Door 1 Friday Start 2 Hour	
63	Door 1 Friday Start 2 Minute	
64	Door 1 Friday End 2 Hour	
65	Door 1 Friday End 2 Minute	
66	Door 1 Saturday Start 1 Hour	
67	Door 1 Saturday Start 1 Minute	
68	Door 1 Saturday End 1 Hour	
69	Door 1 Saturday End 1 Minute	
70	Door 1 Saturday Start 2 Hour	
71	Door 1 Saturday Start 2 Minute	
72	Door 1 Saturday End 2 Hour	
73	Door 1 Saturday End 2 Minute	
74 to ---	Same for Door 2	
--- to ---	Same for Door 3	
--- to N - 2	Same for Door 4	

N - 1	LRC	
N	Footer	Fixed BBh

DOOR STATE EXCEPTIONS (1AH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	1Ah
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Door 1 Enabled Days	0b0XXXXXXX (0b0SMTWTFS)
11	Door 1 Mode Sunday	0h: armed 1h: open 2h: closed
12	Door 1 Mode Monday	
13	Door 1 Mode Tuesday	
14	Door 1 Mode Wednesday	
15	Door 1 Mode Thursday	
16	Door 1 Mode Friday	
17	Door 1 Mode Saturday	
18	Door 1 Sunday Start 1 Hour	
19	Door 1 Sunday Start 1 Minute	
20	Door 1 Sunday End 1 Hour	
21	Door 1 Sunday End 1 Minute	
22	Door 1 Sunday Start 2 Hour	
23	Door 1 Sunday Start 2 Minute	
24	Door 1 Sunday End 2 Hour	
25	Door 1 Sunday End 2 Minute	
26	Door 1 Monday Start 1 Hour	
27	Door 1 Monday Start 1 Minute	
28	Door 1 Monday End 1 Hour	
29	Door 1 Monday End 1 Minute	
30	Door 1 Monday Start 2 Hour	
31	Door 1 Monday Start 2 Minute	
32	Door 1 Monday End 2 Hour	
33	Door 1 Monday End 2 Minute	
34	Door 1 Tuesday Start 1 Hour	
35	Door 1 Tuesday Start 1 Minute	
36	Door 1 Tuesday End 1 Hour	
37	Door 1 Tuesday End 1 Minute	
38	Door 1 Tuesday Start 2 Hour	
39	Door 1 Tuesday Start 2 Minute	
40	Door 1 Tuesday End 2 Hour	
41	Door 1 Tuesday End 2 Minute	
42	Door 1 Wednesday Start 1 Hour	
43	Door 1 Wednesday Start 1 Minute	

44	Door 1 Wednesday End 1 Hour	
45	Door 1 Wednesday End 1 Minute	
46	Door 1 Wednesday Start 2 Hour	
47	Door 1 Wednesday Start 2 Minute	
48	Door 1 Wednesday End 2 Hour	
49	Door 1 Wednesday End 2 Minute	
50	Door 1 Thursday Start 1 Hour	
51	Door 1 Thursday Start 1 Minute	
52	Door 1 Thursday End 1 Hour	
53	Door 1 Thursday End 1 Minute	
54	Door 1 Thursday Start 2 Hour	
55	Door 1 Thursday Start 2 Minute	
56	Door 1 Thursday End 2 Hour	
57	Door 1 Thursday End 2 Minute	
58	Door 1 Friday Start 1 Hour	
59	Door 1 Friday Start 1 Minute	
60	Door 1 Friday End 1 Hour	
61	Door 1 Friday End 1 Minute	
62	Door 1 Friday Start 2 Hour	
63	Door 1 Friday Start 2 Minute	
64	Door 1 Friday End 2 Hour	
65	Door 1 Friday End 2 Minute	
66	Door 1 Saturday Start 1 Hour	
67	Door 1 Saturday Start 1 Minute	
68	Door 1 Saturday End 1 Hour	
69	Door 1 Saturday End 1 Minute	
70	Door 1 Saturday Start 2 Hour	
71	Door 1 Saturday Start 2 Minute	
72	Door 1 Saturday End 2 Hour	
73	Door 1 Saturday End 2 Minute	
74 to ---	Same for Door 2	
--- to ---	Same for Door 3	
--- to N - 2	Same for Door 4	
N - 1	LRC	
N	Footer	Fixed BBh

AREA SETUP (1BH)

RFU command. No need to implement

Byte Index	Data	Value
0	Command Header	Fixed AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W	01h
6	FC	1Ch
7	Response Required? /Response	00h / XXh

8	Payload Length (LSB)	YYh
9	Payload Length(HSB)	XXh
10	Door 1 Area ID	1d to 255d
10	Door 2 Area ID	1d to 255d
11	Door 3 Area ID	1d to 255d
12	Door 4 Area ID	1d to 255d
59	LRC	
60	Footer	Fixed BBh

INPUT DEFINITIONS (1CH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	1Ch
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Input 1 Mapping Address	See List here
11	Input 1 Mode	1h: normally opened 2h: normally closed (fixed mode for door sensor)
12	RFU	0h
13	Input 1 Pulse width LSB	YYh (1d to 65000d)
14	Input 1 Pulse width HSB	XXh (In milliseconds)
15	Input 2 Mapping Address	
16	Input 2 Mode	
17	RFU	
18	Input 2 Pulse width LSB	
19	Input 2 Pulse width HSB	
20	Input 3 Mapping Address	
21	Input 3 Mode	
22	RFU	
23	Input 3 Pulse width LSB	
24	Input 3 Pulse width HSB	
25	Input 4 Mapping Address	
26	Input 4 Mode	
27	RFU	
28	Input 4 Pulse width LSB	
29	Input 4 Pulse width HSB	
30	Input 5 Mapping Address	
31	Input 5 Mode	
32	RFU	
33	Input 5 Pulse width LSB	

34	Input 5 Pulse width HSB	
35	Input 6 Mapping Address	
36	Input 6 Mode	
37	RFU	
38	Input 6 Pulse width LSB	
39	Input 6 Pulse width HSB	
40	Input 7 Mapping Address	
41	Input 7 Mode	
42	RFU	
43	Input 7 Pulse width LSB	
44	Input 7 Pulse width HSB	
45	Input 8 Mapping Address	
46	Input 8 Mode	
47	RFU	
48	Input 8 Pulse width LSB	
49	Input 8 Pulse width HSB	
50	Input 9 Mapping Address	
51	Input 9 Mode	
52	RFU	
53	Input 9 Pulse width LSB	
54	Input 9 Pulse width HSB	
55	Input 10 Mapping Address	
56	Input 10 Mode	
57	RFU	
58	Input 10 Pulse width LSB	
59	Input 10 Pulse width HSB	
60	Input 11 Mapping Address	
61	Input 11 Mode	
62	RFU	
63	Input 11 Pulse width LSB	
64	Input 11 Pulse width HSB	
65	Input 12 Mapping Address	
66	Input 12 Mode	
67	RFU	
68	Input 12 Pulse width LSB	
69	Input 12 Pulse width HSB	
70	LRC	
71	Footer	Fixed BBh

INPUT TO OUTPUT TRIGGERS (1DH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	1Dh
7	Response Required? /Response	00h / XXh

8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Action 1 Input Number (see list here)	
11	Action 1 Output Number (see list here)	
12	Action 1 Relay Action	0h: Nothing 1h: Activate 2h: Deactivate
13	Action 1 Alarm	00h: Disable 01h: Alarm Enable 02h: GPI Reset Enable 03h: Alarm & GPI Reset Enable
14	Action 1 Scope	1h : local 2h : global
15	Action 1 Time LSB	1d to 65533d : for seconds 65534d : up to valid input 65535d : infinite time
16	Action 1 Time HSB	
17 to N - 9	 Action 2 to Action 64	
N - 8	Action 64 Input Number (see list here)	
N - 7	Action 64 Output Number (see list here)	
N - 6	Action 64 Action	
N - 5	Action 64 Alarm Enable?	
N - 4	Action 64 Scope	1h : local 2h : global
N - 3	Action 64 Time LSB	
N - 2	Action 64 Time HSB	
n - 1	LRC	
n	Footer	Fixed BBh

EVENT TO OUTPUT TRIGGERS (1EH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	1Eh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Door Number	1d to 4d
11	Action 1 Event (see list here)	
12	Action 1 Input Number (see list here)	
13	Action 1 Output Number (see list here)	
14	Action 1 Relay Action	0h: Nothing 1h: Open 2h: Close

15	Action 1 Alarm	00h: Disable 01h: Alarm Enable
16	Action 1 Time LSB	1d to 65533d : for seconds 65534d : up to valid input 65535d : infinite time
17	Action 1 Time HSB	
18 to N - 9	 Action 2 to Action 64	
N - 8	Action 64 Input Number (see list here)	
N - 7	Action 64 Output Number (see list here)	
N - 6	Action 64 Action	
N - 5	Action 64 Alarm Enable?	
N - 4	Action 64 Photo Capture Enable?	
N - 3	Action 64 Time LSB	
N - 2	Action 64 Time HSB	
n - 1	LRC	
n	Footer	Fixed BBh

BUZZER LED SETUP (1FH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	1Fh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Allow Internal Buzzer Pulses	1d to 255d
11	Allow Internal Buzzer On LSB	YYh (in milliseconds)
12	Allow Internal Buzzer On HSB	XXh
13	Allow Internal Buzzer Off LSB	
14	Allow Internal Buzzer Off HSB	
15	Deny Internal Buzzer Pulses	
16	Deny Internal Buzzer On LSB	
17	Deny Internal Buzzer On HSB	
18	Deny Internal Buzzer Off LSB	
19	Deny Internal Buzzer Off HSB	
20	Allow External Buzzer Pulses	
21	Allow External Buzzer On LSB	
22	Allow External Buzzer On HSB	
23	Allow External Buzzer Off LSB	
24	Allow External Buzzer Off HSB	
25	Deny External Buzzer Pulses	
26	Deny External Buzzer On LSB	
27	Deny External Buzzer On HSB	
28	Deny External Buzzer Off LSB	
29	Deny External Buzzer Off HSB	

30	Allow Green LED Pulses	
31	Allow Green LED On LSB	
32	Allow Green LED On HSB	
33	Allow Green LED Off LSB	
34	Allow Green LED Off HSB	
35	Deny Green LED Pulses	
36	Deny Green LED On LSB	
37	Deny Green LED On HSB	
38	Deny Green LED Off LSB	
39	Deny Green LED Off HSB	
40	Allow RED LED Pulses	
41	Allow RED LED On LSB	
42	Allow RED LED On HSB	
43	Allow RED LED Off LSB	
44	Allow RED LED Off HSB	
45	Deny RED LED Pulses	
46	Deny RED LED On LSB	
47	Deny RED LED On HSB	
48	Deny RED LED Off LSB	
49	Deny RED LED Off HSB	
50	LRC	
51	Footer	Fixed BBh

DISPLAY & BRIGHTNESS PARAMETERS (20H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	20h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	First Line Message	00h: Nothing
11	Second Line Message	01h: User Name
12	Third Line Message (only for BIOSCRIBE_3S)	02h: Card ID
13	Fourth Line Message (only for BIOSCRIBE_3S)	03h: Employee ID
14	Clock Face (For Bioscribe 3S)	04h: Custom MSZ
15	Hour Format if Digital Face	0h: digital
16	Card ID Display format	1h: analouge (default for bioscribe 3s)
17	Display Picture (DP) On	00h: 12 Hours format (default)
		01h: 24 hour format
		'H' (48h) : Hexadecimal display
		'D' (44h): Decimal Display
		00h: default
		00h: disable
		01h: enable

18 to 49	Access allowed message	Null terminated 32 byte ascii string
50 to 81	Access denied message	Null terminated 32 byte ascii string
82 to 113	Global Message	Null terminated 32 byte ascii string
114	Screen Saver Enable (Bioscribe 3S)	00h: Disable 01h: Enable
115	Screen Saver Timeout	5d to 255d seconds, 0 if disabled
116	Screen Saver change picture time	5d to 255d seconds, 0 if disabled
117	Active brightness level	50d to 100d
118	Idle brightness level	10d to 49d
119	Active to Idle transition time LSB	DDh (1d to 255d * 1000 milliseconds = AABCCDDh)
120	Active to Idle transition time HSB	CCh
121	Active to Idle transition time USB	BBh
122	Active to Idle transition time MSB	AAh
123	Idle to Off transition time LSB	DDh (1d to 255d * 1000 milliseconds = AABCCDDh)
124	Idle to Off transition time HSB	CCh
125	Idle to Off transition time USB	BBh
126	Idle to Off transition time MSB	AAh
127	Active to Idle Transition Enable	0h: disable 1h: enable
128	Idle to Off Transition Enable	0h: disable 1h: enable
129	LRC	
130	Footer	Fixed BBh

SYNC CARD ID FORMAT (21H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	21h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Wigand / Clock Data	1h: Wigand (if 1 then send below bytes) 0h: Clk dta
12	Format 1 Wigand Bits	26d to 254d
13	Format 1 Reader Type	0h: em card 1h: HID Prox 2h: HID I class 3h: MI fare Reverse 4h: MI fare Forward 5h: others

14	Format 1 Site Code (SC) Enable / Disable	0h: disable 1h: enable
15	Format 1 ID start bit	1d to 254d (but < end bit)
16	Format 1 ID end bit	> start bit but <= 254d
17	Format 1 SC start bit position	Bits ranges except ID start & end bits
18	Format 1 SC end bit position	Bits ranges except ID start & end bits
19	Format 1 Direction forward / reverse	0h: forward 1h: reverse
20	Format 1 Parity Enable / Disable	0h: disable 1h: enable
21	Format 1 No. of Parity	1d to 6d
22	Format 1 Parity position (P1)	Parity bit position within bits
23	Format 1 Parity Type (P1)	0h: even 1h: odd
24	Format 1 Parity Calculation start bit (P1)	bit position where to start calculation
25	Format 1 Parity Calculation end bit (P1)	bit position where to end calculation
26	Format 1 Parity position (P2)	
27	Format 1 Parity Type (P2)	
28	Format 1 Parity Calculation start bit (P2)	
29	Format 1 Parity Calculation end bit (P2)	
30	Format 1 Parity position (P3)	
31	Format 1 Parity Type (P3)	
32	Format 1 Parity Calculation start bit (P3)	
33	Format 1 Parity Calculation end bit (P3)	
34	Format 1 Parity position (P4)	
35	Format 1 Parity Type (P4)	
36	Format 1 Parity Calculation start bit (P4)	
37	Format 1 Parity Calculation end bit (P4)	
38	Format 1 Parity position (P5)	
39	Format 1 Parity Type (P5)	
40	Format 1 Parity Calculation start bit (P5)	
41	Format 1 Parity Calculation end bit (P5)	
42	Format 1 Parity position (P6)	
43	Format 1 Parity Type (P6)	
44	Format 1 Parity Calculation start bit (P6)	
45	Format 1 Parity Calculation end bit (P6)	
	Same for Format 2	
	Same for Format 3	
	Same for Format 4	
	Same for Format 5	
N - 1	LRC	
N	Footer	Fixed BBh

OFFLINE FIRMWARE UPDATE (READ FIRMWARE VERSION) (22H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h

5	R/W flag	01h
6	FC	22h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10 to 14	Model ID	4 ascii bytes
15 to --	Model Name	16 ascii bytes
	Current Firmware Version	16 ascii bytes
	Current Firmware Date	16 ascii bytes
	Boot Loader Version	7 ascii bytes
N - 1	LRC	
N	Footer	Fixed BBh

OFFLINE FIRMWARE UPDATE (UPLOAD FIRMWARE INFO) (23H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	23h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Total Length in Bytes LSB	DDh (if size is AABBCDDh)
11	Total Length in Bytes HSB	CCh
12	Total Length in Bytes USB	BBh
13	Total Length in Bytes MSB	AAh
14	Total Segments LSB	DDh (if AABBCDDh)
15	Total Segments HSB	CCh
16	Total Segments USB	BBh
17	Total Segments MSB	AAh
18	CRC16 CCSIT LSB	YYh (if CRC is XYYh)
19	CRC16 CCSIT HSB	XXh
20	LRC	
21	Footer	Fixed BBh

OFFLINE FIRMWARE UPDATE (UPLOAD FIRMWARE DATA) (24H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh

2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	24h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Segment Number LSB	DDh (if number is AABBCDDh)
11	Segment Number HSB	CCh
12	Segment Number USB	BBh
13	Segment Number MSB	AAh
14	Segment Data Length LSB	DDh (if length is AABBCDDh)
15	Segment Data Length HSB	CCh
16	Segment Data Length USB	BBh
17	Segment Data Length MSB	AAh
18 to N - 2	Data bytes (max 1024)	Actual Firmware Data (ascii bytes)
N - 1	LRC	
N	Footer	Fixed BBh

OFFLINE FIRMWARE UPDATE (VERIFY & START UPGRADE) (25H)

This command will Reset Device if previous commands are executed in proper sequence. If any error was there this command will report error

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	25h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	00h
9	Payload length (HSB)	00h
10	LRC	
11	Footer	Fixed BBh

FACTORY RESET (26H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh

3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	26h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	00h
9	Payload length (HSB)	00h
10	LRC	
11	Footer	Fixed BBh

SOFT RESET (RESTART DEVICE) (27H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	27h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	00h
9	Payload length (HSB)	00h
10	LRC	
11	Footer	Fixed BBh

OVER RIDE OUTPUTS (28H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h / 01h
6	FC	28h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Relay 1 Mode BST/BSC/UST = LOCK	0h: armed 1h: open 2h: closed
11	Relay 1 Time LSB	1d to 65533d : for seconds 65535d : infinite

		time
12	Relay 1 Time HSB	
13	Relay 2 Mode BST/BSC/UST = GPO1	
14	Relay 2 Time LSB	
15	Relay 2 Time HSB	
16	Relay 3 Mode	
17	Relay 3 Time LSB	
18	Relay 3 Time HSB	
19	Relay 4 Mode	
20	Relay 4 Time LSB	
21	Relay 5 Time HSB	
22	Relay 5 Mode	
23	Relay 5 Time LSB	
24	Relay 5 Time HSB	
25	Relay 6 Mode	
26	Relay 6 Time LSB	
27	Relay 6 Time HSB	
28	Relay 7 Mode	
29	Relay 7 Time LSB	
30	Relay 7 Time HSB	
31	Relay 8 Mode	
32	Relay 8 Time LSB	
33	Relay 8 Time HSB	
34	LRC	
35	Footer	Fixed BBh

MI-FARE SCAN CSN (29H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	29h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	04h
9	Payload length (HSB)	00h
10	Returned Card ID LSB	
11	Returned Card ID HSB	
12	Returned Card ID USB	
13	Returned Card ID MSB	
14	LRC	
15	Footer	Fixed BBh

MI-FARE WRITE CARD (2AH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	2Ah
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Old key byte 1	
11	Old key byte 2	
12	Old key byte 3	
13	Old key byte 4	
14	Old key byte 5	
15	Old key byte 6	
16	Old Key Type	AAh or BBh
17	New key byte 1	
18	New key byte 2	
19	New key byte 3	
20	New key byte 4	
21	New key byte 5	
22	New key byte 6	
23	New Key Type	
24	Block Number	1d to 255d
25 to 40	Block Data	16 bytes of data
41	LRC	
42	Footer	Fixed BBh

MI-FARE RELEASE CSN (2BH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	2Bh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	00h
9	Payload length (HSB)	00h
10	LRC	
11	Footer	Fixed BBh

MI-FARE SECTOR MODE CONFIG (2CH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	2Ch
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Index/Block number	See Note
11	Block Byte 1Byte	
12	Block Byte 2Byte	
13	Block Byte 3Byte	
14	Block Byte 4Byte	
15	Block Byte 5Byte	
16	Block Byte 6Byte	
17	Block Byte 7Byte	
18	Block Byte 8Byte	
19	Block Byte 9Byte	
20	Block Byte 10Byte	
21	Block Byte 11Byte	
22	Block Byte 12Byte	
23	Block Byte 13Byte	
N - 1	LRC	
N	Footer	Fixed BBh

❖ NOTE FOR COMMAND (2CH) :-

1.SECTOR MODE CONFIGURATION

Index/Block number	(2d)
Block Byte 1Byte	RFU
Block Byte 2Byte	Reader type CSN / Sector-- 1/2

2.EMPLOY ID CONFIGURATION

Index/Block number	(4d)
Block Byte 1Byte	Employee ID On/Of(1/0)
Block Byte 2Byte	Employee Sector
Block Byte 3Byte	Employee Block
Block Byte 4Byte	Employee Byte Address

Block Byte 5Byte	Employee Key Type
Block Byte 6Byte	Employee Key Byte 1
Block Byte /Byte	Employee Key Byte 2
Block Byte 8Byte	Employee Key Byte 3
Block Byte 9Byte	Employee Key Byte 4
Block Byte 10Byte	Employee Key Byte 5
Block Byte 11Byte	Employee Key Byte 6
Block Byte 12Byte	Employee ID length
Block Byte 13Byte	Employee ID format

3.NAME1 CONFIGURATION

Index/Block number	(5d)
Block Byte 1Byte	Name1 On/Of(1/0)
Block Byte 2Byte	Name1 Sector
Block Byte 3Byte	Name1 Block
Block Byte 4Byte	Name1 Byte Address
Block Byte 5Byte	Name1 Key Type
Block Byte 6Byte	Name1 Key Byte 1
Block Byte /Byte	Name1 Key Byte 2
Block Byte 8Byte	Name1 Key Byte 3
Block Byte 9Byte	Name1 Key Byte 4
Block Byte 10Byte	Name1 Key Byte 5
Block Byte 11Byte	Name1 Key Byte 6
Block Byte 12Byte	Name1 ID length

4.NAME2 CONFIGURATION

Index/Block number	(6d)
Block Byte 1Byte	Name2 On/Of(1/0)
Block Byte 2Byte	Name2 Sector
Block Byte 3Byte	Name2 Block
Block Byte 4Byte	Name2 Byte Address
Block Byte 5Byte	Name2 Key Type
Block Byte 6Byte	Name2 Key Byte 1
Block Byte /Byte	Name2 Key Byte 2
Block Byte 8Byte	Name2 Key Byte 3
Block Byte 9Byte	Name2 Key Byte 4
Block Byte 10Byte	Name2 Key Byte 5
Block Byte 11Byte	Name2 Key Byte 6
Block Byte 12Byte	Name2 ID length

5.FP1 CONFIGURATION

Index/Block number	(8D)
Block Byte 1Byte	FP1 On/Of(1/0)
Block Byte 2Byte	FP1 Sector
Block Byte 3Byte	FP1 Block
Block Byte 4Byte	FP1 Byte Address
Block Byte 5Byte	FP1 Key Type
Block Byte 6Byte	FP1 Key Byte 1
Block Byte /Byte	FP1 Key Byte 2
Block Byte 8Byte	FP1 Key Byte 3
Block Byte 9Byte	FP1 Key Byte 4
Block Byte 10Byte	FP1 Key Byte 5

Block Byte 11Byte	FP1 Key Byte 6
Block Byte 12Byte	Fp Size Low Byte (0x01)
Block Byte 13Byte	Fp Size Low Byte (0x40/0x00/0x80)

6.FP2 CONFIGURATION

Index/Block number	(12d)
Block Byte 1Byte	FP2 On/Of(1/0)
Block Byte 2Byte	FP2 Sector
Block Byte 3Byte	FP2 Block
Block Byte 4Byte	FP2 Byte Address
Block Byte 5Byte	FP2 Key Type
Block Byte 6Byte	FP2 Key Byte 1
Block Byte /Byte	FP2 Key Byte 2
Block Byte 8Byte	FP2 Key Byte 3
Block Byte 9Byte	FP2 Key Byte 4
Block Byte 10Byte	FP2 Key Byte 5
Block Byte 11Byte	FP2 Key Byte 6
Block Byte 12Byte	Fp Size Low Byte (0x01)
Block Byte 13Byte	Fp Size Low Byte (0x40/0x00/0x80)

7.DOE CONFIGURATION

Index/Block number	(10d)
Block Byte 1Byte	DOE On/Of(1/0)
Block Byte 2Byte	DOE Sector
Block Byte 3Byte	DOE Block
Block Byte 4Byte	DOE Byte Address
Block Byte 5Byte	DOE Key Type
Block Byte 6Byte	DOE Key Byte 1
Block Byte /Byte	DOE Key Byte 2
Block Byte 8Byte	DOE Key Byte 3
Block Byte 9Byte	DOE Key Byte 4
Block Byte 10Byte	DOE Key Byte 5
Block Byte 11Byte	DOE Key Byte 6
Block Byte 12Byte	DOE Length
Block Byte 13Byte	DOE Format(0/1/2)

8.TRANSACTION ON CARD CONFIGURATION

Index/Block number	(14d)
Block Byte 1Byte	On/Of(1/0)
Block Byte 2Byte	Sector
Block Byte 3Byte	Block
Block Byte 4Byte	Byte Address
Block Byte 5Byte	Key Type
Block Byte 6Byte	Key Byte 1
Block Byte /Byte	Key Byte 2
Block Byte 8Byte	Key Byte 3
Block Byte 9Byte	Key Byte 4
Block Byte 10Byte	Key Byte 5
Block Byte 11Byte	Key Byte 6
Block Byte 12Byte	Length
Block Byte 13Byte	Total Trans (max 100)

9.ACCESS CODE SETTING (

Index/Block number	(21d)
Block Byte 1Byte	On/Off(1/0)
Block Byte 2Byte	Sector
Block Byte 3Byte	Block
Block Byte 4Byte	Byte Address
Block Byte 5Byte	Key Type
Block Byte 6Byte	Key Byte 1
Block Byte /Byte	Key Byte 2
Block Byte 8Byte	Key Byte 3
Block Byte 9Byte	Key Byte 4
Block Byte 10Byte	Key Byte 5
Block Byte 11Byte	Key Byte 6
Block Byte 12Byte	Length
Block Byte 13Byte	Format (Hex:0/ Ascii:1/Decimal:2)

SYNC REASON (2DH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	2Dh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Feature Enable Disable	00h: disable 01h: enable
11	Function Mode	00h: optional , enables menu icon on home screen 01h: Mandatory mode, removes icon form home screen
12	RFU	0h
13		
14	Reason 1 Number	1d to 255d
15	Reason 1 Inout (RFU for now)	00h
16	Reason 1 RFU	00h
17 TO 32	Reason 1 Description	16 char ascii string
33	Reason 1 null	00h
34	Reason 1 Card ID LSB	
35	Reason 1 Card ID HSB	
36	Reason 1 Card ID USB	
37	Reason 1 Card ID MSB	
38	Reason 1 Card EXTID ID LSB (for 64 bit CardID only)	

39	Reason 1 Card EXTID ID HSB	
40	Reason 1 Card EXTID ID USB	
41	Reason 1 Card EXTID ID MSB	
42 to 69	Repeat from 13 to 40 for Reason 2 (28 bytes)	
71 to 97	Repeat from 13 to 40 for Reason 3 (28 bytes)	
98 to 125	Repeat from 13 to 40 for Reason 4 (28 bytes)	
126 to 153	Repeat from 13 to 40 for Reason 5 (28 bytes)	
154 to 181	Repeat from 13 to 40 for Reason 6 (28 bytes)	
182 to 209	Repeat from 13 to 40 for Reason 7 (28 bytes)	
210 to 237	Repeat from 13 to 40 for Reason 8 (28 bytes)	
238	LRC	
239	Footer	Fixed BBh

PAUSE / RESUME TRANSACTIONS (2EH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	2Eh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	01h
9	Payload length (HSB)	00h
10	Action	0h: resume 1h: pause
11	LRC	
12	Footer	Fixed BBh

START TRANSACTION RECOVERY (2FH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	2Fh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Unique Sequence Number LSB	4byte unique id

11	Unique Sequence Number HSB	
12	Unique Sequence Number USB	
13	Unique Sequence Number MSB	
14	Start Year	00d to 99d (for 2000 to 2099)
15	Start Month	01d to 12d
16	Start Date	01d to 31d (keep validation for leap)
17	Start Hour	00d to 23d
18	Start Minute	00d to 59d
19	Start Second	00d to 59d
20	End Year	00d to 99d (for 2000 to 2099)
21	End Month	01d to 12d
22	End Date	01d to 31d (keep validation for leap)
23	End Hour	00d to 23d
24	End Minute	00d to 59d
25	End Second	00d to 59d
26	LRC	
27	Footer	Fixed BBh

CLEAR TRANSACTIONS (30H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	30h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	CID LSB	
11	CID HSB	
12	MAC ID 1	
13	MAC ID 2	
14	MAC ID 3	
15	MAC ID 4	
16	MAC ID 5	
17	MAC ID 6	
18	LRC	
19	Footer	Fixed BBh

DELETE ALL RULES / SPECIFIC RULE(31H)

Byte Index	Data	Value
0	Header	AAh

1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	31h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	02h
9	Payload length (HSB)	00h
10	Rule Number LSB	YYh (XXYYh = Rule Number) (send FFFFh to delete all rules)
11	Rule Number HSB	XXh
12	LRC	
13	Footer	Fixed BBh

RESET RULE STATE (32H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	32h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Scope	0h: All Users 1h: Particular User
11	Rule Number LSB	Rule Number to Reset
12	Rule Number MSB	
13	UID LSB	
14	UID HSB	
15	UID USB	
16	UID MSB	
17	EXTD ID LSB (for 64 bit CardID only)	
18	EXTD ID HSB	
19	EXTD ID USB	
20	EXTD ID MSB	
21	LRC	
22	Footer	Fixed BBh

MASTER SLAVE SETUP (33H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	32h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Device Mode	0h: Master Mode 1h: Slave Mode
11	Master IPv4 address segment 1	C0h (if ip is 192.100.100.90)
12	Master IPv4 address segment 2	64h
13	Master IPv4 address segment 3	64h
14	Master IPv4 address segment 4	5Ah
15	Master Port LSB	
16	Master Port HSB	
17	RFU	
18	LRC	
19	Footer	Fixed BBh

RULE SETUP (NORMAL / SIMPLE APB) (34H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	34h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	Rule Strictness Entry	0h : disabled 1h : hard 2h : soft
13	Rule Strictness Exit	0h : disabled 1h : hard 2h : soft
14	Daily Reset Enable	0h: Disable

25	Door 4	
26	Entry To Entry Time LSB	DDh (0d to 86400d = AABCCDDh)
27	Entry To Entry Time HSB	CCh
28	Entry To Entry Time USB	BBh
29	Entry To Entry Time MSB	AAh
30	Entry To Exit Time LSB	DDh (0d to 86400d = AABCCDDh)
31	Entry To Exit Time HSB	CCh
32	Entry To Exit Time USB	BBh
33	Entry To Exit Time MSB	AAh
34	Exit To Entry Time LSB	DDh (0d to 86400d = AABCCDDh)
35	Exit To Entry Time HSB	CCh
36	Exit To Entry Time USB	BBh
37	Exit To Entry Time MSB	AAh
38	Exit To Exit Time LSB	DDh (0d to 86400d = AABCCDDh)
39	Exit To Exit Time HSB	CCh
40	Exit To Exit Time USB	BBh
41	Exit To Exit Time MSB	AAh
42	LRC	
43	Footer	Fixed BBh

RULE SETUP (CONDITIONAL APB) (36H)

Not Implemented. RFU

RULE SETUP (CONCENTRIC CIRCLE) (37H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	37h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	Rule Strictness Entry	0h : disabled 1h : hard 2h : soft
13	Rule Strictness Exit	0h : disabled 1h : hard 2h : soft
14	Daily Reset Enable	0h: Disable 1h: Reset on Specified Time 2h: Reset on GPI Trigger 3h: Reset on Time or GPI
15	Daily Reset Time Hour	0d to 23d

16	Daily Reset Time Minute	0d to 59d
17	Reset GPI Number	1h to 12h
18	User Type LSB	Bit wise encoded + or ed together 000000000000000000000000000000001b: for type 1 10000000000000000000000000000000b: for type 32
19	User Type HSB	
20	User Type USB	
21	User Type MSB	
22	Door 1 layer Number	0h: rule does not apply on this door 1h, 2h, 3h layer number
23	Door 2 layer Number	
24	Door 3 layer Number	
25	Door 4 layer Number	
26	LRC	
27	Footer	Fixed BBh

RULE SETUP (MATRIX APB) (38H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	38h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	Entry Strictness	1h : hard 2h : soft
13	Exit Strictness	1h : hard 2h : soft
14	RFU	0h
15	RFU	0h
16	RFU	0h
17	RFU	0h
18	User Type LSB	Bit wise encoded + or ed together 000000000000000000000000000000001b: for type 1 100000000000000000000000000000000b: for type 32
19	User Type HSB	
20	User Type USB	
21	User Type MSB	
22	Door 1 Group Number	0d: rule does not apply on this door 1d to 8d: group number 255d: this is a reset door
23	Door 2 Group Number	
24	Door 3 Group Number	
25	Door 4 Group Number	
26	Rule Reset After Minutes LSB	0d: reset functionality disabled 1d to 65535d minutes
27	Rule Reset After Minutes HSB	
28	Minimum Time In Reset Gate LSB	0d: reset functionality disabled 1d to 65535d minutes
29	Minimum Time In Reset Gate HSB	
30	Row 1 Column1 access (1 st cell on	0bAABBCCDD (1 byte) where

	table)	AA = Exit to Entry access BB = Entry to Exit access (RFU) CC = Exit to Exit access. (RFU) DD = Entry to Entry access.(RFU) Possible values. 00b: NULL 01b: Allow 10b: Deny
31	Row 1 Column2 access (2 nd cell on table)	
32 to ...	Cell 3 rd to cell 63 access	
N - 2	Row 8 Column8 access (64 th cell on table)	
N - 1	LRC	
N	Footer	Fixed BBh

RULE SETUP (ACCESS LIMIT) (39H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	39h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	Rule Strictness Entry	1h : hard 2h : soft
13	RFU	0h
14	Daily Reset Enable	0h: Disable 1h: Enable
15	Daily Reset Time Hour	0d to 23d
16	Daily Reset Time Minute	0d to 59d
17	Local / Global	0h: Local Individual Door Wise rule 1h: Global Group Wise Rule (MS Required)
18	User Type LSB	Bit wise encoded + or ed together 000000000000000000000000000000001b: for type 1 100000000000000000000000000000000b: for type 32
19	User Type HSB	
21	User Type USB	
22	User Type MSB	
23	Door 1	0h: disable 1h: enable
24	Door 2	
25	Door 3	
26	Door 4	
27	Access Limit LSB	0h: Disable on this door 1h to 65000h limit
28	Access Limit HSB	
29	LRC	
30	Footer	Fixed BBh

RULE SETUP (ABSENT USER) (3AH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	3Ah
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	Rule Strictness Entry	0h: disabled 1h : hard 2h : soft
13	RFU	0h
14	RFU	0h
15	RFU	0h
16	RFU	0h
17	Local / Global	0h: Local Individual Door Wise rule 1h: Global Group Wise Rule (MS Required)
18	User Type LSB	Bit wise encoded + or ed together 000000000000000000000000000000001b: for type 1 100000000000000000000000000000000b: for type 32
19	User Type HSB	
20	User Type USB	
21	User Type MSB	
22	Door 1	0h: disable 1h: enable
23	Door 2	
24	Door 3	
25	Door 4	
26	Absent Days	1d to 250d
27	LRC	
28	Footer	Fixed BBh

RULE SETUP (CUSTODIAN) (3BH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	3Bh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh

9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	Rule Strictness Entry	0h: disabled 1h : hard 2h : soft
13	RFU	0h
14	Daily Reset Enable	0h: Disable 1h: Enable
15	Daily Reset Time Hour	0d to 23d
16	Daily Reset Time Minute	0d to 59d
17	RFU	0h
18	User Type LSB	Bit wise encoded + or ed together 000000000000000000000000000000001b: for type 1 100000000000000000000000000000000b: for type 32
19	User Type HSB	
20	User Type USB	
21	User Type MSB	
22	Door 1	0h: disable 1h: enable
23	Door 2	
24	Door 3	
25	Door 4	
26	Custodian 1 UID LSB	DDh (If card id is AABBCDDh)
27	Custodian 1 UID HSB	CCh
28	Custodian 1 UID USB	BBh
29	Custodian 1 UID MSB	AAh
30	Custodian 1 EXTD ID LSB (for 64 bit CardID only)	DDh (If card id is AABBCDDh)
31	Custodian 1 EXTD ID HSB	CCh
32	Custodian 1 EXTD ID USB	BBh
33	Custodian 1 EXTD ID MSB	AAh
34	Custodian 2 UID LSB	
35	Custodian 2 UID HSB	
36	Custodian 2 UID USB	
37	Custodian 2 UID MSB	
38	Custodian 2 EXTD ID LSB (for 64 bit CardID only)	
39	Custodian 2 EXTD ID HSB	
40	Custodian 2 EXTD ID USB	
41	Custodian 2 EXTD ID MSB	
42	Custodian 3 UID LSB	
45	Custodian 3 UID HSB	
46	Custodian 3 UID USB	
47	Custodian 3 UID MSB	
48	Custodian 3 EXTD ID LSB (for 64 bit CardID only)	
49	Custodian 3 EXTD ID HSB	
50	Custodian 3 EXTD ID USB	
51	Custodian 3 EXTD ID MSB	
52	Custodian 4 UID LSB	
53	Custodian 4 UID HSB	
54	Custodian 4 UID USB	
55	Custodian 4 UID MSB	
56	Custodian 4 EXTD ID LSB (for 64 bit CardID only)	
57	Custodian 4 EXTD ID HSB	

58	Custodian 4 EXTD ID USB	
59	Custodian 4 EXTD ID USB	
60	Custodian 5 UID LSB	
61	Custodian 5 UID HSB	
62	Custodian 5 UID USB	
63	Custodian 5 UID MSB	
64	Custodian 5 EXTD ID LSB (for 64 bit CardID only)	
65	Custodian 5 EXTD ID HSB	
66	Custodian 5 EXTD ID USB	
67	Custodian 5 EXTD ID MSB	
68	LRC	
69	Footer	Fixed BBh

RULE SETUP (MULTIPLE OCCUPANCY) (3CH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	3Ch
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	Rule Strictness Entry	0h : disabled 1h : hard 2h : soft
13	Rule Strictness Exit	0h : disabled 1h : hard 2h : soft
14	RFU	0h
15	RFU	0h
16	RFU	0h
17	Local / Global	0h: Local Individual Door Wise rule 1h: Global Group Wise Rule (MS Required)
18	User Type LSB	Bit wise encoded + or ed together 000000000000000000000000000000001b: for type 1 10000000000000000000000000000000b: for type 32
19	User Type HSB	
20	User Type USB	
21	User Type MSB	
22	Door 1	0h: disable 1h: enable
23	Door 2	
24	Door 3	
25	Door 4	
26	Maximum Count LSB	YYh (0d to 65000d) (0d for no limit)
27	Maximum Count HSB	XXh

28	Minimum Count LSB	0d to 255d (0d for no limit)
29	LRC	
30	Footer	Fixed BBh

RULE SETUP (SENTRY CHECK) (3DH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	3Dh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	RFU	0h
13	RFU	0h
14	RFU	0h
14	RFU	0h
15	RFU	0h
16	RFU	0h
17	User Type LSB	Bit wise encoded + or ed together 000000000000000000000000000000001b: for type 1 10000000000000000000000000000000b: for type 32
18	User Type HSB	
19	User Type USB	
20	User Type MSB	
21	Door 1	0h: disable 1h: enable
22	Door 2	
23	Door 3	
24	Door 4	
25	Period Minutes (LSB)	YYh (where XXYy = 1d to 1440d minutes)
26	Period Minutes (HSB)	XXh
27	Access Type	1h: allow access 2h: do not allow access
28	Minimum Man Alive	1d to 5d
29	Dead Man 1 UID LSB	
30	Dead Man 1 UID HSB	
31	Dead Man 1 UID USB	
32	Dead Man 1 UID MSB	
33	Dead Man 1 EXTD ID LSB (for 64 bit CardID only)	
34	Dead Man 1 EXTD ID HSB	
35	Dead Man 1 EXTD ID USB	
36	Dead Man 1 EXTD ID MSB	
37	Dead Man 2 UID LSB	
38	Dead Man 2 UID HSB	

39	Dead Man 2 UID USB	
40	Dead Man 2 UID MSB	
41	Dead Man 2 EXTD ID LSB (for 64 bit CardID only)	
42	Dead Man 2 EXTD ID HSB	
43	Dead Man 2 EXTD ID USB	
44	Dead Man 2 EXTD ID MSB	
45	Dead Man 3 UID LSB	
46	Dead Man 3 UID HSB	
47	Dead Man 3 UID USB	
48	Dead Man 3 UID MSB	
49	Dead Man 3 EXTD ID LSB (for 64 bit CardID only)	
50	Dead Man 3 EXTD ID HSB	
51	Dead Man 3 EXTD ID USB	
52	Dead Man 3 EXTD ID MSB	
53	Dead Man 4 UID LSB	
54	Dead Man 4 UID HSB	
55	Dead Man 4 UID USB	
56	Dead Man 4 UID MSB	
57	Dead Man 4 EXTD ID LSB (for 64 bit CardID only)	
58	Dead Man 4 EXTD ID HSB	
59	Dead Man 4 EXTD ID USB	
60	Dead Man 4 EXTD ID MSB	
61	Dead Man 5 UID LSB	
62	Dead Man 5 UID HSB	
63	Dead Man 5 UID USB	
64	Dead Man 5 UID MSB	
65	Dead Man 5 EXTD ID LSB (for 64 bit CardID only)	
66	Dead Man 5 EXTD ID HSB	
67	Dead Man 5 EXTD ID USB	
68	Dead Man 5 EXTD ID MSB	
69	LRC	
70	Footer	Fixed BBh

RULE SETUP (WEB POST) (3EH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	3Eh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh

10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	Rule Strictness Entry	0h : disabled 1h : hard 2h : soft
13	Rule Strictness Exit	0h : disabled 1h : hard 2h : soft
14	RFU	0h
15	RFU	0h
16	RFU	0h
17	RFU	1h
18	User Type LSB	Bit wise encoded + or ed together 000000000000000000000000000000001b: for type 1 10000000000000000000000000000000b: for type 32
19	User Type HSB	
20	User Type USB	
21	User Type MSB	
22	Payload format	0h: HTTP Post (JSON) 1h: ASCII (JSON only, no http header) 2h: Raw Binary
23	Online Verification Needed?	0h: no 1h: yes (server should response with “True”/”False” ASCII string case sensitive. In case of RAW format 1h/0h
24	Server IPv4 Segment 1	
25	Server IPv4 Segment 2	
26	Server IPv4 Segment 3	
27	Server IPv4 Segment 4	
28	Server Port LSB	
29	Server Port HSB	
30	SSL enable on this port	0h: disable 1h: enable
31	Info Field LSB	Bit wise or ed together 000000000000000000000000000000001b: Card ID 0000000000000000000000000000000010b: Site Code 00000000000000000000000000000000100b: CID 000000000000000000000000000000001000b: Date Time 0000000000000000000000000000000010000b: Reader Number 00000000000000000000000000000000100000b: Gate Number 000000000000000000000000000000001000000b: Access Info
32	Info Field HSB	
33	Info Field USB	
34	Info Field MSB	
35 to N - 2	HTTP api path	255d ASCII string zero padded
N - 1	LRC	
N	Footer	Fixed BBh

RULE SETUP (INTERLOCKING) (3FH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h

5	R/W flag	01h
6	FC	3Fh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	RFU	0h
13	RFU	0h
14	RFU	0h
15	RFU	0h
16	RFU	0h
17	No Of Doors In this Rule	1h to 8h
18	RFU	0h
19	RFU	
20	RFU	
21	With Switch / Switch Less	Bit wise 0b : With Switch / 1b : Switch Less Example: ABCDEFGHb where (1 byte) A : Door 8 B : Door 7 C : Door 6 D : Door 5 E : Door 4 F : Door 3 G : Door 2 H : Door 1
22	Door 1 Index	0h : disable
23	Door 2 Index	1h to 8h : valid Indies from DIT matrix
24	Door 3 Index	
25	Door 4 Index	
26	Other Door State On Door 1 Open	Bit wise + or ed together Valid values for each bit 0b : unlocked 1b: locked Example: ABCDEFGHb where (1 byte) A : Door 8 B : Door 7 C : Door 6 D : Door 5 E : Door 4 F : Door 3 G : Door 2 H : Door 1
27	Other Door State On Door 2 Open	
28	Other Door State On Door 3 Open	
29	Other Door State On Door 4 Open	
30	Other Door State On Door 5 Open	
31	Other Door State On Door 6 Open	
32	Other Door State On Door 7 Open	
33	Other Door State On Door 8 Open	
34	Delay after Door 1 Close	Bit wise + or ed together Valid values for each bit 0b : no 1b: yes Example: ABCDEFGHb where (1 byte) A : Door 8 B : Door 7 C : Door 6 D : Door 5 E : Door 4 F : Door 3 G : Door 2
35	Delay after Door 2 Close	
36	Delay after Door 3 Close	
37	Delay after Door 4 Close	
38	Delay after Door 5 Close	
39	Delay after Door 6 Close	
40	Delay after Door 7 Close	
41	Delay after Door 8 Close	

		H : Door 1
42	Buzzer / LED Indication While Door 1 Open	Bit wise + or ed together
43	Buzzer / LED Indication While Door 2 Open	Valid values for each bit
44	Buzzer / LED Indication While Door 3 Open	0b : no 1b: yes
45	Buzzer / LED Indication While Door 4 Open	Example: ABCDEFGHb where (1 byte)
46	Buzzer / LED Indication While Door 5 Open	A : Door 8
47	Buzzer / LED Indication While Door 6 Open	B : Door 7
48	Buzzer / LED Indication While Door 7 Open	C : Door 6
49	Buzzer / LED Indication While Door 8 Open	D : Door 5
		E : Door 4
		F : Door 3
		G : Door 2
		H : Door 1
50	Door 1 Delay Time LSB	YYh (where YYXXh ranges from 0d to 65000d)
51	Door 1 Delay Time HSB	XXh
52	Door 2 Delay Time LSB	
53	Door 2 Delay Time HSB	
54	Door 3 Delay Time LSB	
55	Door 3 Delay Time HSB	
56	Door 4 Delay Time LSB	
57	Door 4 Delay Time HSB	
58	Door 5 Delay Time LSB	
59	Door 5 Delay Time HSB	
60	Door 6 Delay Time LSB	
61	Door 6 Delay Time HSB	
62	Door 7 Delay Time LSB	
63	Door 7 Delay Time HSB	
64	Door 8 Delay Time LSB	
65	Door 8 Delay Time HSB	
66	Fail Safe / Fail Secure?	0h: Fail Secure mechanism 1h: Fail Safe Mechanism
67	Fail Safe Time LSB	YYh (where YYXXh ranges from 0d to 65000d)
68	Fail Safe Time HSB	XXh
69	LRC	
70	Footer	Fixed BBh

RULE SETUP (GROUP AUTHENTICATION) (40H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	40h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)

11	Rule Number (HSB)	XXh
12	Rule Strictness Entry	0h : disabled 1h : hard
13	Rule Strictness Exit	0h : disabled 1h : hard
14	RFU	0h
15	RFU	0h
16	RFU	0h
17	User Type LSB	Bit wise encoded + or ed together 000000000000000000000000000000001b: for type 1 100000000000000000000000000000000b: for type 32
18	User Type HSB	
19	User Type USB	
20	User Type MSB	
21	Door 1 Entry Authentication mode	0h: Single / Do not apply 1h: Dual Authentication 2h: Triple Authentication
22	Door 1 Exit Authentication	
23	Door 2 Entry Authentication	
24	Door 2 Exit Authentication	
25	Door 3 Entry Authentication	
26	Door 3 Exit Authentication	
27	Door 4 Entry Authentication	
28	Door 4 Exit Authentication	
29	Group Enable/Disable	Enable: 0x01 / Disable: 0x00
30	Door 1 Entry Group No1 LSB	
31	Door 1 Entry Group No1 HSB	
32	Door 1 Entry Group No2 LSB	
33	Door 1 Entry Group No2 HSB	
34	Door 1 Entry Group No3 LSB	
35	Door 1 Entry Group No3 HSB	
36	Door 1 Exit Group No1 LSB	
37	Door 1 Exit Group No1 HSB	
38	Door 1 Exit Group No2 LSB	
39	Door 1 Exit Group No2 HSB	
40	Door 1 Exit Group No3 LSB	
41	Door 1 Exit Group No3 HSB	
42	Door 2 Entry Group No1 LSB	
43	Door 2 Entry Group No1 HSB	
44	Door 2 Entry Group No2 LSB	
45	Door 2 Entry Group No2 HSB	
46	Door 2 Entry Group No3 LSB	
47	Door 2 Entry Group No3 HSB	
48	Door 2 Exit Group No1 LSB	
49	Door 2 Exit Group No1 HSB	
50	Door 2 Exit Group No2 LSB	
51	Door 2 Exit Group No2 HSB	
52	Door 2 Exit Group No3 LSB	
53	Door 2 Exit Group No3 HSB	
54	Door 3 Entry Group No1 LSB	
55	Door 3 Entry Group No1 HSB	
56	Door 3 Entry Group No2 LSB	
57	Door 3 Entry Group No2 HSB	
58	Door 3 Entry Group No3 LSB	
59	Door 3 Entry Group No3 HSB	

60	Door 3 Exit Group No1 LSB	
61	Door 3 Exit Group No1 HSB	
62	Door 3 Exit Group No2 LSB	
63	Door 3 Exit Group No2 HSB	
64	Door 3 Exit Group No3 LSB	
65	Door 3 Exit Group No3 HSB	
66	Door 4 Entry Group No1 LSB	
67	Door 4 Entry Group No1 HSB	
68	Door 4 Entry Group No2 LSB	
69	Door 4 Entry Group No2 HSB	
70	Door 4 Entry Group No3 LSB	
71	Door 4 Entry Group No3 HSB	
72	Door 4 Exit Group No1 LSB	
73	Door 4 Exit Group No1 HSB	
74	Door 4 Exit Group No2 LSB	
75	Door 4 Exit Group No2 HSB	
76	Door 4 Exit Group No3 LSB	
77	Door 4 Exit Group No3 HSB	
78	LRC	
79	Footer	Fixed BBh

RULE SETUP (MULTIPLE AUTHENTICATION) (40H) (DEPRECATED, DO NOT USE)

[illegible]

18	User Type USB	10000000000000000000000000000000b: for type 32
19	User Type MSB	
20	Door 1 Entry Authentication mode	0h: Single 1h: Double 2h: Triple
21	Door 1 Exit Authentication	
22	Door 2 Entry Authentication	
23	Door 2 Exit Authentication	
24	Door 3 Entry Authentication	
25	Door 3 Exit Authentication	
26	Door 4 Entry Authentication	
27	Door 4 Exit Authentication	
28	LRC	
29	Footer	Fixed BBh

GET FINGER TEMPLATE (41H)

Read mode will get only local templates (from sd card), will not get templates from fp module. Returned data length will depend upon finger size setting in fppara. (nominally 384 bytes per finger template)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	0Ch
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Number Of Templates Required	1h or 2h
11	UID LSB	
12	UID HSB	
13	UID USB	
14	UID MSB	
15	EXTD ID LSB (for 64 bit CardID only)	
16	EXTD ID HSB	
17	EXTD ID USB	
18	EXTD ID MSB	
19	Template 1 Index	0h to 9h
20	Template 2 Index	0h to 9h
21	Returned Template 1 size LSB	80h (if 384d = 180h)
22	Returned Template 1 size HSB	01h
23.. to M	Returned Template 1 Data byte 0	
	...	
M	Returned Template 1 Data byte last	
M+1	Returned Template 2 size LSB	
M+2	Returned Template 2 size HSB	
M+3	Returned Template 2 Data byte 0	
...	...	

N - 2	Returned Template 2 Data byte last	
N - 1	LRC	
N	Footer	Fixed BBh

DELETE SPECIFIC PARAMETERS (42H)

This command may take time if multiple parameters are being deleted

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	42h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Delete All Users + User Exceptions + Fingers	0h: no 1h: yes
11	Delete All Access Tables (TZ, Wekkoff, Holiday, TZ grp)	
12	Delete Other Para (GPI, Door Exceptions, Door mode, Input Actions, Event Actions, Rules, APB)	
13	RFU	
14	RFU	
15	RFU	
16	RFU	
17	RFU	
18	LRC	
19	Footer	Fixed BBh

USER ANTI PASS-BACK OVERRIDE (43H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	43h
7	Response Required? /Response	00h / XXh

8	Payload length (LSB)	18h
9	Payload length (HSB)	00h
10	UID LSB	
11	UID HSB	
12	UID USB	
13	UID MSB	
14	Site Code LSB	
15	Site Code HSB	
16	Site Code USB	
17	Site Code MSB	
18	Set or Delete on Door 1	0h: no change
19	Set or Delete on Door 2	1h: set
20	Set or Delete on Door 3	2h: delete
21	Set or Delete on Door 4	
	What to Override	1h: ALL APB 2h: Access Limit 4h: Absent user 8h: Custodian 10h: Occupancy 20h: Multiple Auth
22	RFU	0h
23	LRC	
24	Footer	Fixed BBh

USER AUTHENTICATION CHANGE / FINGER BYPASS (44H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	44h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	09h
9	Payload length (HSB)	00h
10	UID LSB	
11	UID HSB	
12	UID USB	
13	UID MSB	
14	Site Code LSB	
15	Site Code HSB	
16	Site Code USB	
17	Site Code MSB	
18	Enable Bypass on Door 1	1h: enable
19	Enable Bypass on Door 2	0h: disable
20	Enable Bypass on Door 3	
21	Enable Bypass on Door 4	

22	Entry Authentication Mode Door 1	01h: Card Only
23	Exit Authentication Mode Door 1	02h: Card + Pin
24	Entry Authentication Mode Door 2	03h: Card / Pin
25	Exit Authentication Mode Door 2	04h: Card + Finger
26	Entry Authentication Mode Door 3	05h: Card + Pin + Finger
27	Exit Authentication Mode Door 3	06h: Only Finger
28	Entry Authentication Mode Door 4	07h: Card / Finger
29	Exit Authentication Mode Door 4	
30	LRC	
31	Footer	Fixed BBh

USER CUSTOM MESSAGE (45H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	45h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	09h
9	Payload length (HSB)	00h
10	UID LSB	
11	UID HSB	
12	UID USB	
13	UID MSB	
14	Site Code LSB	
15	Site Code HSB	
16	Site Code USB	
17	Site Code MSB	
18	Start Year	
19	Start Month	
20	Start Date	
21	RFU	
22	RFU	
23	End Year	
24	End Month	
25	End Data	
26	RFU	
27	RFU	
28	Custom Message	32 characters ascii string
29	LRC	
30	Footer	Fixed BBh

NTP PARAMETERS (46H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	46h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	NTP client on off	0h: disabled 1h: enabled
11	Time zone Sign	1d: negative 2d: positive
12	Time zone hour	0d to 12d
13	Time zone minute	0d to 59d
14	NTP server address	32 char ascii string. Must be valid web address excluding 'Http://' or 'Https://'
14 + 32	LRC	
14 + 33	Footer	Fixed BBh

CAMERA CAPTURE ON SOFTWARE EVENTS (47H) (VARIABLE LENGTH COMMAND)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	46h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Capture with GPI 1 alarm	0h: no 1h: yes
11	Capture with GPI 2 alarm	
12	RFU	
13	RFU	
14	Capture with Temper 1 Alarm	
15	Capture with Temper 2 Alarm	Software Event code. See Event list
16	Capture with This Event 1	
17	Capture with This Event 2	
18	Capture with This Event 3	
19	Capture with This Event ...	
...	...	
N - 2		
N - 1	LRC	
N	Footer	Fixed BBh

DEVICE PASSWORD RESET (48H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	48h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	09h
9	Payload length (HSB)	00h
10	Pass Char 1	Ascii string max 8 char
11	Pass Char 2	Alpha numeric for BIOSCRIBE 3S
12	Pass Char 3	Numeric only for TwinXS 3S
13	Pass Char 4	
14	Pass Char 5	
15	Pass Char 6	

16	Pass Char 7	
17	Pass Char 8	
18	Blank Byte	00h
19	Admin Card ID 1 LSB	DDh (if id is AABBCDDh)
20	Admin Card ID 1 HSB	CCh
21	Admin Card ID 1 USB	BBh
22	Admin Card ID 1 MSB	AAh
23	Admin Card ID 2 LSB	
24	Admin Card ID 2 HSB	
25	Admin Card ID 2 USB	
26	Admin Card ID 2 MSB	
27	Admin Card ID 3 LSB	
28	Admin Card ID 3 HSB	
29	Admin Card ID 3 USB	
30	Admin Card ID 3 MSB	
31	Admin Card ID 4 LSB	
32	Admin Card ID 4 HSB	
33	Admin Card ID 4 USB	
34	Admin Card ID 4 MSB	
35	Admin Card ID 5 LSB	
36	Admin Card ID 5 HSB	
37	Admin Card ID 5 USB	
38	Admin Card ID 5 MSB	
39	LRC	
40	Footer	Fixed BBh

BLE PARA (49H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	49h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	BLE access enable?	00h: disabled 01h: enabled
11	Tx Power Level	0d to 127d
12	Key Byte 1	
13	Key Byte 2	
14	Key Byte 3	
15	Key Byte 4	
16	Key Byte 5	
17	Key Byte 6	
18 to 41	Door Name	24 byte ascii string.. if name is shorter than 24 bytes

		pad with hex 00h
42	LRC	
43	Footer	Fixed BBh

SET FINGER ENCRYPTION (4AH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	2Dh
2	Total Length (HSB)	00h
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	4Ah
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	21h
9	Payload length (HSB)	00h
10	Fingerprint template encryption enable?	00h: disabled 01h: enabled
11 to 43	32 byte AES Key	32 bytes key
44	LRC	
45	Footer	Fixed BBh

ADD EXTENDED USER INFORMATION (4BH)

Note: This command may be expanded in future to incorporate more information.

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	4Bh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	UID LSB	DDh (if card id is AABCCDDh)
11	UID HSB	CCh
12	UID USB	BBh
13	UID MSB	AAh
14	EXTD ID LSB (for 64 bit CardID only)	DDh (if card id is AABCCDDh)
15	EXTD ID HSB	CCh

16	EXTD ID USB	BBh
17	EXTD ID MSB	AAh
18	Update Escort	UpdateEscort: 0x01 otherwise 0x00
19	Escort group ID LSB Door 1	2 Byte group ID for Escort Rule, numeric or alpha numeric
20	Escort group ID HSB Door 1	
21	Escort group ID LSB Door 2	
22	Escort group ID HSB Door 2	
23	Escort group ID LSB Door 3	
24	Escort group ID HSB Door 3	
25	Escort group ID LSB Door 4	
26	Escort group ID HSB Door 4	
27	Update Multiple Auth group	Update Multiple Auth group: 0x01 otherwise 0x00
28	Multiple Auth group1 ID LSB	2 Byte group ID for Multiple Auth Rule
29	Multiple Auth group1 ID HSB	
30	Multiple Auth group2 ID LSB	
31	Multiple Auth group2 ID HSB	
32	Multiple Auth group3 ID LSB	
33	Multiple Auth group3 ID HSB	
34	Multiple Auth group4 ID LSB	
35	Multiple Auth group4 ID HSB	
36	LRC	
37	Footer	Fixed BBh

ENABLE / DISABLE ETHERNET MODULES (F9H)

Upon executing this command in write mode device will reboot.

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	11h
2	Total Length (HSB)	00h
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	F9h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	05h
9	Payload length (HSB)	00h
10	FTP module	00h: Disable 01h: Enable
11	HTTP web module	
12	TCP Modbus Module	
13	SNMP Module	
14	BLE Module	
15	LRC	

16	Footer	Fixed BBh
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SOUND & VOLUME PARAMETERS (4CH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	10h
2	Total Length (HSB)	00h
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	4Ch
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	04h
9	Payload length (HSB)	00h
10	Key Volume	0d to 10 d 01h: Beep style (for allow it will play 1 long beep, for deny it will play 3 short beep) 02h: Voice prompt
11	Notification Volume	
12	Allow volume style	
13	Deny volume style	
14	LRC	
15	Footer	Fixed BBh

USER PASSWORD (PIN) CHANGE (4DH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	19h
2	Total Length (HSB)	00h
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	4Dh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	0Dh
9	Payload length (HSB)	00h
10	UID LSB	78h(If UID is 0x12345678)
11	UID HSB	56h
12	UID USB	34h
13	UID MSB	12h
14	EXTD ID LSB (for 64 bit CardID only)	78h(If EXTD ID is 0x12345678) (Currently all zeros)
15	EXTD ID HSB	56h
16	EXTD ID USB	34h
17	EXTD ID MSB	12h
18	PIN LSB	00h (set all zeros in this field to reset pin, else set pin)
19	PIN HSB	00h

30 to 125	Server Name	Null terminated ASCII chars (96 MAX)
126 to 221	Publish Topic Name	Null terminated ASCII chars (96 MAX)
222 to 245	MQTT User Name	Null terminated ASCII chars (24 MAX)
246 to 269	MQTT User Password	Null terminated ASCII chars (24 MAX)
270 to 293	MQTT Client ID	Null terminated ASCII chars (24 MAX)
294	Field 1 Code	00h: Field Disable 01h: Card ID 02h: Employee Name 03h: Employee ID 04h: CID 05h: Date & Time 06h: Access Code 07h: In / Out 08h: Reader Number 09h: Keypad Input
295 to 309	Field 1 Name	Null terminated ASCII chars (14 MAX)
310	Field 1 Null	00h
311	Field 2 Code	
312 to 325	Field 2 Name	
326	Field 2 Null	
327	Field 3 Code	
328 to 341	Field 3 Name	
342	Field 3 Null	
343	Field 4 Code	
And so on	Field 4 Name	
	Field 4 Null	
	Field 5 Code	
	Field 5 Name	
	Field 5 Null	
	Field 6 Code	
	Field 6 Name	
	Field 6 Null	
	Field 7 Code	
	Field 7 Name	
	Field 7 Null	
	Field 8 Code	
	Field 8 Name	
	Field 8 Null	
422	Keypad Input Enable	00h: disable 01h: Enable
423	Keypad Input Type	00h: Numeric 01h: Alpha Numeric
424	Keypad Input length	1d to 8d
425	RFU	00h
426	RFU	00h
427	RFU	00h
428	RFU	00h
429	RFU	00h
430 to 517	RFU	00h
518	LRC	
519	Footer	Fixed BBh

TEMPERATURE DETECTION PARAMETERS (50H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	1Ch
2	Total Length (HSB)	00h
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	50h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	10h
9	Payload length (HSB)	00h
10	Temperature Reader Enabled	0h: No 01h: Yes
11	Temperature Threshold Integer part	25h (37d.80d = 25h.50h) (37.80 °Celsius is default)
12	Temperature Threshold Fraction Part	50h
13	Temperature Threshold Check	00h: Disable 01h: Enable
14	RFU	00h
15	RFU	00h
16	RFU	00h
17	RFU	00h
18	RFU	00h
19	RFU	00h
20	RFU	00h
21	RFU	00h
22	RFU	00h
23	RFU	00h
24	RFU	00h
25	RFU	00h
26	LRC	
27	Footer	Fixed BBh

ARM DISARM MODE SETTINGS (51H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	10h
2	Total Length (HSB)	00h
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h

6	FC	51h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	04h
9	Payload length (HSB)	00h
10	DOOR 1	00h: Door Armed 01h: Door Disarmed
11	DOOR 2	
12	DOOR 3	
13	DOOR 4	
14	LRC	
15	Footer	Fixed BBh

MIFARE MADCONFIG (52H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	52h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
11	Application Index	1d to 8d
12	This is MAD Config	00h (Fix)
13	Enable	00h: Disable 01h: Enable
14	App ID LSB	YYh (if APP id is XXYYh)
15	App ID MSB	XXh
16	MAD Key Type	AAh or BBh
17	MAD Key Byte 1	
18	MAD Key Byte 2	
19	MAD Key Byte 3	
20	MAD Key Byte 4	
21	MAD Key Byte 5	
22	MAD Key Byte 6	
16	App Key Type	AAh or BBh
17	App Key Byte 1	
18	App Key Byte 2	
19	App Key Byte 3	
20	App Key Byte 4	
21	App Key Byte 5	
22	App Key Byte 6	
23	Skip Validation?	00h: No 01h: Yes

24	LRC	
25	Footer	Fixed BBh

MIFARE MAD PARAMETERS(52H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	52h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
11	Application Index	1d to 8d
12	MAD Parameters	01h: Access Code 02h: Employee Name 03h: Employee ID 04h: Date of Start 05h: Date of Expiry 06h: FP1 07h: FP2
13	Enable?	00h: disabled 01h: Enabled
14	Data Offset LSB	YYh (if Offset = XXYYh)
15	Data Offset MSB	XXh
16	Data Length LSB	YYh (if Length = XXYYh)
17	Data Length MSB	XXh
18	Data Format	00h: Hex Format 01h: ASCII Format 02h: BCD Format
19	LRC	
20	Footer	Fixed BBh

OSDP CONTROLLER SETUP(53H)

0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	53h
7	Response Required? /Response	00h / XXh

8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	OSDP Enable	1: OSDP Enable 0: OSDP disable
11	Port LSB	Lower byte of port number
12	Port MSB	Higher byte of port number
13	RFU	
14	RFU	
15	RFU	
16	RFU	
17	RFU	
18	RFU	
19	Reader 1 type	0:disable 1: RS485 2 : IP/Ethernet
20	RFU	
21	RFU	
22	RFU	
23	Reader IPv4 address segment 1	00 in case of RS485 otherwise IP address
24	Reader IPv4 address segment 2	00 in case of RS485 otherwise IP address
25	Reader IPv4 address segment 3	00 in case of RS485 otherwise IP address
26	Reader IPv4 address segment 4	00 in case of RS485 otherwise IP address
27	Reader 2 type	0:disable 1: RS485 2 : IP/Ethernet
28	RFU	
29	RFU	
30	RFU	
31	Reader IPv4 address segment 1	00 in case of RS485 otherwise IP address
32	Reader IPv4 address segment 2	00 in case of RS485 otherwise IP address
33	Reader IPv4 address segment 3	00 in case of RS485 otherwise IP address
34	Reader IPv4 address segment 4	00 in case of RS485 otherwise IP address
35	Reader 3 type	0:disable 1: RS485 2 : IP/Ethernet
36	rfu	
37	RFU	
38	RFU	
39	Reader IPv4 address segment 1	00 in case of RS485 otherwise IP address
40	Reader IPv4 address segment 2	00 in case of RS485 otherwise IP address
41	Reader IPv4 address segment 3	00 in case of RS485 otherwise IP address
42	Reader IPv4 address segment 4	00 in case of RS485 otherwise IP address
43	Reader 4 type	0:disable 1: RS485 2 : IP/Ethernet
44	RFU	
45	RFU	
46	RFU	
47	Reader IPv4 address segment 1	
48	Reader IPv4 address segment 2	
49	Reader IPv4 address segment 3	
50	Reader IPv4 address segment 4	
51	LRC	

53	Footer	Fixed BBh

OSDP READER SETUP(54H)

0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	54h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	OSDP Enable	0h:OSDP mode disable 1h:OSDP mode enable
11	Master IP address byte 1	
12	Master IP address byte 2	
13	Master IP address byte 3	
14	Master IP address byte 4	
15	Port LSB	
16	Port MSB	
17	Address (1 to 4) , Reader ID	1 to 4
18	isFailSafe	0h: Fail Secure 1h: Fail Safe
19	RFU	
20	RFU	
21	RFU	
22	RFU	
23	LRC	
24	Footer	Fixed BBh

EAC READER SETUP(55H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	55h
7	Response Required? /Response	00h / XXh

8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Reader Number(1 to 4)	XXh
11	ELSGW IPv4 address segment 1	C0h (if ip is 192.100.100.90)
12	ELSGW IPv4 address segment 2	64h
13	ELSGW IPv4 address segment 3	64h
14	ELSGW IPv4 address segment 4	5Ah
15	Multicast IPv4 address segment 1	C0h (if ip is 192.100.100.101)
16	Multicast IPv4 address segment 2	64h
17	Multicast IPv4 address segment 3	64h
18	Multicast IPv4 address segment 4	65h
19	Port Number LSB	YYh
20	Port Number MSB	XXh
21	ELSGW Address LSB	XXh
22	Receiver Address LSB	XXh
23	Verification Type LSB	XXh
24	Verification Location LSB	XXh
25	Boarding Floor LSB	XXh
26	Boarding Floor HSB	YYh
27	Boarding F/R LSB	XXh
28	Destination F/R LSB	XXh
29	Terminal Number LSB	If (Terminal number is 0x1234 then) 0x12
30	Terminal Number HSB	0x34
31	LRC	
32	Footer	Fixed BBh

RULE SETUP SERVER AUTH(56H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	56h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Rule Number (LSB)	YYh (1d to 65000d)
11	Rule Number (HSB)	XXh
12	Rule Strictness Entry	1h : hard
13	Rule Strictness Exit	2h : soft
14	RFU	
15	RFU	
16	RFU	
17	RFU	

18	RFU	
19	RFU	
21	RFU	
22	RFU	
23	Door 1	0h: disable 1h: enable
24	Door 2	
25	Door 3	
26	Door 4	
27	Apply rule on Entry	0h: no 1h: yes
28	Apply rule on Exit	
29	Skip User Search	
30	Skip User Validation	
31	RFU	
32	RFU	
33	RFU	
34	RFU	
29	LRC	
30	Footer	Fixed BBh

EVENT ACTION EXTENDED(57H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	57h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Door Number	1h to 4h
11	Event 1 Delay LSB	XXYYh = 1d to 65533d where
12	Event 1 Delay HSB	
13	Event 1 RFU	
14	Event 1 RFU	
15	Event 1 RFU	
16	Event 1 RFU	
17	Event 1 RFU	
18 + 7	Same for Event 2	
	Repeat 11 to 17 for 64 times	
N-2	Same for Event 64	
N - 1	LRC	
N	Footer	Fixed BBh

INPUT ACTION EXTENDED(58H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	58h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Door Number	1h to 4h
11	Event 1 Delay LSB	XXYYh = 1d to 65533d where
12	Event 1 Delay HSB	
13	Event 1 RFU	
14	Event 1 RFU	

15	Event 1 RFU	
16	Event 1 RFU	
17	Event 1 RFU	
18 + 7	Same for Event 2	
	Repeat 11 to 17 for 64 times	
N-2	Same for Event 64	
N - 1	LRC	
N	Footer	Fixed BBh

CONFIGURABLE OUTPUT CONTROL (59H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h / 01h
6	FC	59h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Relay 1 Mode BSC/UST = LOCK	0h: Normally ON 1h: Normally OFF
11	Relay 1 Time LSB	1 ~ 1000S LSB
12	Relay 1 Time HSB	1 ~ 1000S HSB
13	Relay 2 Mode BSC/UST = GPO1	0h: Normally ON 1h: Normally OFF
14	Relay 2 Time LSB	1 ~ 1000S LSB
15	Relay 2 Time HSB	1 ~ 1000S HSB
16	RFU	
17	RFU	
18	RFU	
19	RFU	
20	RFU	
21	RFU	
22	RFU	
23	RFU	
24	RFU	
25	RFU	
26	RFU	
27	RFU	
28	RFU	
29	RFU	
30	LRC	
31	FOOTER	BBh

SERVER AUTH QUERY(D1H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	D1h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	QueryID LSB	Per device Unique ID for each query.
11	QueryID HLSB	
12	QueryID USB	
13	QueryID MSB	
14	Rule number LSB	
15	Rule number HSB	
16	Door ID	1d to 4d
17	InOut	0h: null 1h: in 2h: out
18	CardID LSB	
19	CardID HLSB	
21	CardID USB	
22	CardID MSB	
23	Extd CardID LSB	
24	Extd CardID HLSB	
25	Extd CardID USB	
26	Extd CardID MSB	
27	Hour	
28	Minute	
29	Second	
30	Date	
31	Month	
32	Year	
33	LRC	
34	Footer	Fixed BBh

SERVER AUTH RESPONSE(D1H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	02h
4	RFU	01h
5	R/W flag	01h
6	FC	D1h

7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	QueryID LSB	Per device Unique ID for each query.
11	QueryID HLSB	
12	QueryID USB	
13	QueryID MSB	
14	AllowDenyNoSay	0h: no say, continue as per device auth
15		1h: forcefully allow
		2h: deny
16	Deny Reason	16 byte ascii string
17	RFU	0h
18	LRC	
19	Footer	Fixed BBh

CANTEEN MANAGEMENT SYSTEM COMMANDS

ITEM DATABASE (C0H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	C0h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Item Number (LSB)	(1d to 512d)
11	Item Number (HSB)	
12	Active / Inactive	1 : Active 0 : Inactive
13	Item Price (LSB)	(1d to 1000000d) If item price is 100 Rs, then price field should be 10000d (2710h) in Paisa. Item Price (LSB) = 10h; Item Price (HSB) = 27h; Item Price (USB) = 00h; Item Price (MSB) = 00h;
14	Item Price (HSB)	
15	Item Price (USB)	
16	Item Price (MSB)	
17 + 16	Item Name	16 Bytes in ASCII string.
33	Week Day Group	Bit wise encoded + or ed together 00000001b: for Sunday 01000000b: for Saturday
34	Canteen Code	
35	Vendor Code	1d to 127d
36	LRC	
37	Footer	Fixed BBh

ITEM DELETE (C1H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	C1h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Item Number (LSB)	(1d to 512d) for single item delete FFFFh (65535d) for All items delete
11	Item Number (HSB)	
12	LRC	
13	Footer	Fixed BBh

ADD ITEM GROUP(C2H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	C2h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Item Group Number	1d to 99d
11	Location ID	
12	Item Number (LSB)	(1d to 512d)
13	Item Number (HSB)	
14	Timezone1 Number	1d to 127d
15	Timezone2 Number	1d to 127d
16	Timezone3 Number	1d to 127d
17	Timezone4 Number	1d to 127d
18	Item Restriction Type	0 : Unlimited 1 : Timezone wise 2 : Daily 3 : Weekly 4 : Monthly
19	Item Restriction Quantity	1d to 99d
20	Subsidize Price (LSB)	(1d to 1000000d) If subsidize price is 100 Rs, then price field should be
21	Subsidize Price (HSB)	
22	Subsidize Price (USB)	

23	Subsidize Price (MSB)	10000d (2710h) in Paisa.
24	Subsidize Restriction Type	0 : Unlimited 1 : Timezone wise 2 : Daily 3 : Weekly 4 : Monthly
25	Subsidize Restriction Quantity	1d to 99d
26	RFU	
27	RFU	
28	RFU	
29	RFU	
30	RFU	
31	RFU	
32	RFU	
33	RFU	
34	RFU	
35	RFU	
36	RFU	
37	RFU	
38	RFU	
39	RFU	
40	RFU	
41	LRC	
42	Footer	Fixed BBh

GROUP ITEM DELETE (C3H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	C3h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Item Group Number	1d to 99d : for delete specific item from Group. 255d : for delete all groups
10	Item Number (LSB)	(1d to 512d) : for single item delete.
11	Item Number (HSB)	FFFFh : for delete all items from specified group.
12	LRC	
13	Footer	Fixed BBh

VENDOR DB (C4H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	C4h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Vendor ID	0d to 127d
11 to N-2	Vendor Name	32 Characters in ASCII string
N-1	LRC	
N	Footer	Fixed BBh

PRINT SLIP CONFIGURATION (C5H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	C5h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Print Slip	00h : Disable 01h : Enable
11	Line Feed	0d to 9d
12	Auto Cutter Enabled	0d: Disable 1d: Enable
13	Auto Cutter Command Byte 1	
14	Auto Cutter Command Byte 2	
15	Print Slip Field 1	01h: Company Header
16	Print Slip Field 2	02h: Employee Name
17	Print Slip Field 3	03h: Card ID
18	Print Slip Field 4	04h: Employee ID
19	Print Slip Field 5	05h: Vendor Name
20	Print Slip Field 6	06h: Date & Time
21	Print Slip Field 7	07h: Item & Quantity
22	Print Slip Field 8	08h: Transaction Amount

23	Print Slip Field 9	09h: Remaining Balance
24	Print Slip Field 10	0ah: Previous Balance
25	Print Slip Field 11	0bh: Auxiliary Header1
26	Print Slip Field 12	0ch: Auxiliary Header2
27	Print Slip Field 13	0dh: Canteen Code
28	Print Slip Field 14	0eh: Controller ID
29	Print Slip Field 15	
30	LRC	
31	Footer	Fixed BBh

CANTEEN PARAMETERS (C6H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	C6h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Multiple Item Selection	0 : Disable
11	Multiple Quantity Selection	1 : Enable
12	Master Slave Configuration	0 : Stand Alone 1 : Validation from Server
13	RFU	
14	RFU	
15	RFU	
16	LRC	
17	Footer	Fixed BBh

PRINT HEADER/INDEX DATA (C7H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	C7h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Company Header / Auxiliary Index	0 : Company Header 1 : Auxiliary Index1 2 : Auxiliary Index2
11	Data Byte1	
...		
...		
42	Data Byte32	
43	LRC	
44	Footer	Fixed BBh

QUERY TO MASTER (C8H) , MASTER SLAVE COMMUNICATION

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	C8h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	00h
9	Payload length (HSB)	00h
10	Card ID (LSB)	
11	Card ID (HSB)	
12	Card ID (USB)	
13	Card ID (MSB)	
14	Extended ID (LSB)	
15	Extended ID (HSB)	
16	Extended ID (USB)	
17	Extended ID (MSB)	
18	Current Date	
19	Current Month	

20	Current Year	
21	Current Hour	
22	Current Minute	
23	Current Second	
24	Item1 Number LSB	
25	Item1 Number MSB	
26	Item1 Quantity	
27	Item1 Group Number	
28	Item1 Canteen Code	
29	Item1 Vendor Code	
30	Item2 Number LSB	
31	Item2 Number MSB	
32	Item2 Quantity	
33	Item2 Group Number	
34	Item2 Canteen Code	
35	Item2 Vendor Code	
36	Item3 Number LSB	
37	Item3 Number MSB	
38	Item3 Quantity	
39	Item3 Group Number	
40	Item3 Canteen Code	
41	Item3 Vendor Code	
42	Item4 Number LSB	
43	Item4 Number MSB	
44	Item4 Quantity	
45	Item4 Group Number	
46	Item4 Canteen Code	
47	Item4 Vendor Code	
48	Item5 Number LSB	
49	Item5 Number MSB	
50	Item5 Quantity	
51	Item5 Group Number	
52	Item5 Canteen Code	
53	Item5 Vendor Code	
54	LRC	
55	Footer	Fixed BBh

RESPONSE FROM MASTER (C8H) , MASTER SLAVE COMMUNICATION

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	C8h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	00h
9	Payload length (HSB)	00h

10	Card ID (LSB)	
11	Card ID (HSB)	
12	Card ID (USB)	
13	Card ID (MSB)	
14	Site Code (LSB)	
15	Site Code (HSB)	
16	Site Code (USB)	
17	Site Code (MSB)	
18	Item1 Number LSB	
19	Item1 Number MSB	
20	Item1 Price LSB	
21	Item1 Price HSB	
22	Item1 Price USB	
23	Item1 Price MSB	
24	Item1 Quantity	
25	Item1 Error Code	01h : Successful XXh : Error Code (See below Table)
26	Item2 Number LSB	
27	Item2 Number MSB	
28	Item2 Price LSB	
29	Item2 Price HSB	
30	Item2 Price USB	
31	Item2 Price MSB	
32	Item2 Quantity	
33	Item2 Error Code	01h : Successful XXh : Error Code (See below Table)
34	Item3 Number LSB	
35	Item3 Number MSB	
36	Item3 Price LSB	
37	Item3 Price HSB	
38	Item3 Price USB	
39	Item3 Price MSB	
40	Item3 Quantity	
41	Item3 Error Code	01h : Successful XXh : Error Code (See below Table)
42	Item4 Number LSB	
43	Item4 Number MSB	
44	Item4 Price LSB	
45	Item4 Price HSB	
46	Item4 Price USB	
47	Item4 Price MSB	
48	Item4 Quantity	
49	Item4 Error Code	01h : Successful XXh : Error Code (See below Table)
50	Item5 Number LSB	
51	Item5 Number MSB	
52	Item5 Price LSB	
53	Item5 Price HSB	
54	Item5 Price USB	
55	Item5 Price MSB	
56	Item5 Quantity	
57	Item5 Error Code	01h : Successful XXh : Error Code (See below Table)
58	Transaction Balance (LSB)	

69	Transaction Balance (HSB)	
60	Transaction Balance (USB)	
61	Transaction Balance (MSB)	
62	User Remaining Balance (LSB)	
63	User Remaining Balance (HSB)	
64	User Remaining Balance (USB)	
65	User Remaining Balance (MSB)	
66	User Previous Balance (LSB)	
67	User Previous Balance (HSB)	
68	User Previous Balance (USB)	
69	User Previous Balance (MSB)	
70	LRC	
71	Footer	Fixed BBh

CANTEEN MASTER/SLAVE COMMUNICATION ERROR CODES

Byte Index	Data
1d	Successful
101d	Daily Item Limit Exceeded
102d	Weekly Item Limit Exceeded
103d	Monthly Item Limit Exceeded
104d	Timezone Wise Item Limit Exceeded
105d	Balance Limit Exceeded

PUSH COMMAND CODES (DEVICE TO SERVICE)

CONNECTION IDENTIFIER(E0)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	25h (37d)
2	Total Length (HSB)	00h
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	E0h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	19h (25d)
9	Payload length (HSB)	00h
10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	CID LSB	YYh (if CID is XXYYh)
17	CID HSB	XXh
18	Communication Interface	1h: ETH 2h: WIFI 3h: GSM
19 to 28	Protocol Version Char 1 to 10	16 Char ASCII string
29	Current Year	
30	Current Month	
31	Current Date	
32	Current Hour	
33	Current Minute	
34	Current Second	
35	LRC	
36	Footer	Fixed BBh

PUSH TRANSACTIONS (E1H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	E1h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh

9	Payload length (HSB)	XXh
10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	No of transactions in this frame	1d to 32d
17	T1 UID LSB	
18	T1 UID HSB	
19	T1 UID USB	
20	T1 UID MSB	
21	T1 SC LSB	
22	T1 SC HSB	
23	T1 SC USB	
24	T1 SC MSB	
25	T1 CID LSB	
26	T1 CID HSB	
27	T1 Transaction Type	50h: 'P' punch Transaction 48h: 'H' hard alarm 53h: 'S' soft alarm 47h: 'G' gps data 43h: 'C' Canteen data
28	T1 Year	0dto 99d
29	T1 Month	1d to 12d
30	T1 Date	1d to 31d
31	T1 Hour	0d to 23d
32	T1 Minute	0d to 59d
33	T1 Second	0d to 59d
34	T1 Access Info / Input Number	See Access Events / Input numbers
35	T1 Reader Number	1d to 8d
36	T1 Door Number	1d to 4d
37	T1 RFU	0h
38	T1 In punch / out punch	0d: null 1d: in 2d: out
39	T1 Reason Flag	0h: no reason, any other value (see reason codes)
40	T1 Is Photo available	0d: no photo captured 1d: photo captured
41	T1 Trans ID LSB	
42	T1 Trans ID HSB	
43	T1 Trans ID USB	
44	T1 Trans ID MSB	
45 to ...	T2 data	
... to ...	T3 data	
... to N-2	T4 to T32 data	
N - 1	LRC	
N	Footer	Fixed BBh

PUSH PHOTO HEADER (E2H) (DEPRECATED)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	E2h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	Image ID LSB	
17	Image ID HSB	
18	Image ID USB	
19	Image ID MSB	
20	Image Size in Byte LSB	
21	Image Size in Byte HSB	
22	Image Size in Byte USB	
23	Image Size in Byte MSB	
24	Total Chunks LSB	
25	Total Chunks HSB	
26	Last Chunk Size LSB	
27	Last Chunk Size HSB	
28	CID LSB	
29	CID HSB	
30	LRC	
31	Footer	Fixed BBh

PUSH PHOTO DATA (E3H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	E3h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh

10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	CID LSB	YYh (if CID is XXYh)
17	CID HSB	XXh
18	Image ID LSB	DDh (if AABBCDDh))
19	Image ID HSB	CCh
20	Image ID USB	BBh
21	Image ID MSB	AAh
22	Total Chunks LSB	DDh (if AABBCDDh))
23	Total Chunks HSB	CCh
24	Total Chunks USB	BBh
25	Total Chunks MSB	AAh
26	Current Chunk LSB	DDh (if AABBCDDh))
27	Current Chunk HSB	CCh
28	Current Chunk USB	BBh
29	Current Chunk MSB	AAh
30 to N - 2	Image Data	Jpg file data
N - 1	LRC	
N	Footer	Fixed BBh

PUSH HARD ALARMS (E4H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	E4h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	No of transactions in this frame	1d to 32d
17	RFU	
18	RFU	
19	RFU	
20	RFU	

21	RFU	
22	RFU	
23	RFU	
24	RFU	
25	T1 CID LSB	
26	T1 CID HSB	
27	T1 Transaction Type	50h: 'P' punch Transaction 48h: 'H' hard alarm 53h: 'S' soft alarm 47h: 'G' gps data 43h: 'C' Canteen data
28	T1 Year	0dto 99d
29	T1 Month	1d to 12d
30	T1 Date	1d to 31d
31	T1 Hour	0d to 23d
32	T1 Minute	0d to 59d
33	T1 Second	0d to 59d
34	Input Number	Input numbers
35	RFU	1d to 8d
36	RFU	1d to 4d
37	RFU	1d to 4d
38	RFU	0d: in 1d: out
39	RFU	0h: no reason, any other value <u>see reason codes</u>
40	T1 Is Photo available	0d: no photo captured 1d: photo captured
41	T1 Trans ID LSB	
42	T1 Trans ID HSB	
43	T1 Trans ID USB	
44	T1 Trans ID MSB	
45 to ...	T2 data	
... to ...	T3 data	
... to N-2	T4 to T32 data	
N - 1	LRC	
N	Footer	Fixed BBh

PUSH SOFT ALARMS (E5H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	E5h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh

10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	No of transactions in this frame	1d to 32d
17	T1 UID LSB	
18	T1 UID HSB	
19	T1 UID USB	
20	T1 UID MSB	
21	T1 SC LSB	
22	T1 SC HSB	
23	T1 SC USB	
24	T1 SC MSB	
25	T1 CID LSB	
26	T1 CID HSB	
27	T1 Transaction Type	50h: 'P' punch Transaction 48h: 'H' hard alarm 53h: 'S' soft alarm 47h: 'G' gps data 43h: 'C' Canteen data
28	T1 Year	0dto 99d
29	T1 Month	1d to 12d
30	T1 Date	1d to 31d
31	T1 Hour	0d to 23d
32	T1 Minute	0d to 59d
33	T1 Second	0d to 59d
34	T1 Access Info / Input Number	See Access Events / Input numbers
35	T1 Reader Number	1d to 8d
36	T1 Door Number	1d to 4d
37	T1 Area number	1d to 4d
38	T1 In punch / out punch	0d: in 1d: out
39	T1 Reason Flag	0h: no reason, any other value <u>see reason codes</u>
40	T1 Is Photo available	0d: no photo captured 1d: photo captured
41	T1 Trans ID LSB	
42	T1 Trans ID HSB	
43	T1 Trans ID USB	
44	T1 Trans ID MSB	
45 to ...	T2 data	
... to ...	T3 data	
... to N-2	T4 to T32 data	
N - 1	LRC	
N	Footer	Fixed BBh

PUSH GPS DATA (E6H)

PUSH CANTEEN DATA (E7H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	E7h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	No of transactions in this frame	1d to 32d
17	T1 UID LSB	
18	T1 UID HSB	
19	T1 UID USB	
20	T1 UID MSB	
21	T1 SC LSB	
22	T1 SC HSB	
23	T1 SC USB	
24	T1 SC MSB	
25	T1 CID LSB	
26	T1 CID HSB	
27	T1 Transaction Type	43h: 'C' Canteen data
28	T1 Year	0dto 99d
29	T1 Month	1d to 12d
30	T1 Date	1d to 31d
31	T1 Hour	0d to 23d
32	T1 Minute	0d to 59d
33	T1 Second	0d to 59d
34	T1 Access Info	
35	T1 Reader Number	1d to 8d
36	T1 Door Number	1d to 4d
37	T1 RFU	0h
38	T1 In punch / out punch	0d: null 1d: in 2d: out
39	T1 Reason Flag	0h: no reason, any other value (see reason codes)
40	T1 Is Photo available	0d: no photo captured

		1d: photo captured
41	T1 Trans ID LSB	
42	T1 Trans ID HSB	
43	T1 Trans ID USB	
44	T1 Trans ID MSB	
45	T1 Item Number LSB	
46	T1 Item Number MSB	
47	T1 Item Quantity	
48	T1 Item Price LSB	
49	T1 Item Price HSB	
50	T1 Item Price USB	
51	T1 Item Price MSB	
42 to ...	T2 data	
... to ...	T3 data	
... to N-2	T4 to T32 data	
N - 1	LRC	
N	Footer	Fixed BBh

RECOVERY TRANSACTION FRAME (E8H)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	E7h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	Unique Sequence Number LSB	4byte unique id
17	Unique Sequence Number HSB	
18	Unique Sequence Number USB	
19	Unique Sequence Number MSB	
20	No. of Transactions in this frame	1d to 32d
21	T1 Transaction type	5h: recovered punch transactions
22	T1 CID LSB	
23	T1 CID HSB	
24	T1 UID LSB	
25	T1 UID HSB	
26	T1 UID USB	
27	T1 UID MSB	

28	T1 SC LSB	
29	T1 SC HSB	
30	T1 SC USB	
31	T1 SC MSB	
32	T1 Year	0d to 99d
33	T1 Month	1d to 12d
34	T1 Date	1d to 31d
35	T1 Hour	0d to 23d
36	T1 Minute	0d to 59d
37	T1 Second	0d to 59d
38	T1 Access Info / Input Number	See Access Events / Input numbers
39	T1 Reader Number	1d to 8d
40	T1 Door Number	1d to 4d
41	T1 Area number	1d to 4d
42	T1 In punch / out punch	0d: in 1d: out
43	T1 Is Photo available	0d: no photo captured 1d: photo captured
44	T1 Trans ID LSB	
45	T1 Trans ID HSB	
46	T1 Trans ID USB	
47	T1 Trans ID MSB	
48 to ---	T2 data	
--- to ---	T3 data	
--- to N - 2	T4 to T32 data	
N - 1	LRC	
N	Footer	Fixed BBh

RECOVERY REPORT FRAME (E9H)

Legend:

Missed: not uploaded, missed transaction found during recovery

Uploaded: uploaded transaction found during recovery

Garbage: useless / corrupt transaction found during recovery

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	E8h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh

10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	Unique Sequence Number LSB	4byte unique id
17	Unique Sequence Number HSB	
18	Unique Sequence Number USB	
19	Unique Sequence Number MSB	
20	Error Flag	1h: recovery Success & finished 2h: end date not found 3h: start data not found
21	Missed LSB	DDh (if AABBCDDh) CCh BBh AAh (for internal purpose only.. do not show this info on screen)
22	Missed HSB	
23	Missed USB	
24	Missed MSB	
25	Uploaded LSB	
26	Uploaded HSB	
27	Uploaded USB	
28	Uploaded MSB	
29	Garbage LSB	
30	Garbage HSB	
31	Garbage USB	
32	Garbage MSB	
33	LRC	
34	Footer	Fixed BBh

GET PENDING EVENT LIST (EAH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	0Fh
2	Total Length (HSB)	00h
3	Command Or Response	01h / 02h
4	RFU	01h
5	R/W flag	00h
6	FC	EAh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	04h
9	Payload length (HSB)	00h
10	Pending LSB	DDh (if AABBCDDh) CCh BBh AAh (Where AABBCDDh = no. of pending events)
11	Pending HSB	
12	Pending USB	
13	Pending MSB	
14	LRC	
15	Footer	Fixed BBh

NEW TRANSACTION FRAME (EBH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	EBh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	No. of transactions in this frame	1d to 32d
17	Size of each Transaction	31d / 1Fh
18	T1 UID LSB	
19	T1 UID HSB	
20	T1 UID USB	
21	T1 UID MSB	
22	T1 SC LSB	
23	T1 SC HSB	
24	T1 SC USB	
25	T1 SC MSB	
26	T1 CID LSB	
27	T1 CID HSB	
28	T1 Transaction Type	50h: 'P' punch Transaction 48h: 'H' hard alarm 53h: 'S' soft alarm 47h: 'G' gps data 43h: 'C' Canteen data
29	T1 Year	0dto 99d
30	T1 Month	1d to 12d
31	T1 Date	1d to 31d
32	T1 Hour	0d to 23d
33	T1 Minute	0d to 59d
34	T1 Second	0d to 59d
35	T1 Access Info / Input Number	See Access Events / Input numbers
36	T1 Reader Number	1d to 8d
37	T1 Door Number	1d to 4d
38	T1 RFU	0h
39	T1 In punch / out punch	0d: null

		1d: in 2d: out
40	T1 Reason Flag	0h: no reason, any other value (see reason codes)
41	T1 Is Photo available	0d: no photo captured 1d: photo captured
42	T1 Trans ID LSB	
43	T1 Trans ID HSB	
44	T1 Trans ID USB	
45	T1 Trans ID MSB	
46	T1 Temperature Integer part	XXh (if temp = XX.YY Degree Celsius)
47	T1 temperature Fraction Part	YYh
48	T1 RFU	
49 to ...	T2 data	
... to ...	T3 data	
... to N-2	T4 to T32 data	
N - 1	LRC	
N	Footer	Fixed BBh

PUSH LOG (ECH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	ECh
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	MAC ID 1	11h (if mac id is 11:22:33:44:55:66)
11	MAC ID 2	22h
12	MAC ID 3	33h
13	MAC ID 4	44h
14	MAC ID 5	55h
15	MAC ID 6	66h
16	Admin UID LSB	
17	Admin UID HSB	
18	Admin UID USB	
19	Admin UID MSB	
20	Year	0dto 99d
21	Month	1d to 12d

22	Date	1d to 31d
23	Hour	0d to 23d
24	Minute	0d to 59d
25	Second	0d to 59d
26	RFU	
27	RFU	
28	RFU	
29	RFU	
30 to N-2	Log Data	ASCII String, max 235 bytes
N - 1	LRC	
N	Footer	Fixed BBh

QUERY COMMANDS

GENERAL QUERY TO SERVER (DOH)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	D0h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Query Type	01d: Check Balance / Bill
11	UID LSB	78h(If UID is 0x12345678)
12	UID HSB	56h
13	UID USB	34h
14	UID MSB	12h
15 to n-2	Null terminated response, present only in response frame from server.	Max 1023 ASCII char. Each lines must be saperated by '\n'. last byte must be 00h.
n-1	LRC	
n	Footer	Fixed BBh

SERVER AUTHENTICATION QUERY / RESPONSE (D1H)

Query Frame (From Device)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	00h
6	FC	D1h
7	Response Required? /Response	00h / XXh
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Auth ID LSB	78h (if auth id is 12345678h)
11	Auth ID HSB	56h
12	Auth ID USB	34h
13	Auth ID MSB	12h
14	Rule Number LSB	YYh (where rule number is XYYh)
15	Rule Number HSB	HHh

16	Door Number	1h2h,3h or 4h
17	InOut	1h: In 2h: Out 0h: Null
18	UID LSB	Card ID
19	UID HSB	
20	UID USB	
21	UID MSB	
22	Extd LSB	Extended Card ID (for cards more then 32 bits) RFU as of now
23	Extd HSB	
24	Extd USB	
25	Extd MSB	
26	Hour	01d to 23d
27	Minute	00d to 59d
28	Second	00d to 59d
N - 1	LRC	
N	Footer	Fixed BBh

Response Frame (From Server)

Byte Index	Data	Value
0	Header	AAh
1	Total Length (LSB)	YYh
2	Total Length (HSB)	XXh
3	Command Or Response	01h
4	RFU	01h
5	R/W flag	01h
6	FC	D1h
7	Response Required? /Response	01h //send success always here
8	Payload length (LSB)	YYh
9	Payload length (HSB)	XXh
10	Auth ID LSB	78h (if auth id is 12345678h)
11	Auth ID HSB	56h
12	Auth ID USB	34h
13	Auth ID MSB	12h
14	Result	00h: No Say 01h: Allowed 02h: Denied
15 + 16	Deny Reason	16 byte ASCII encoded deny string (optional, send all zeros if not available)
32	Fixed zero	00h
N - 1	LRC	
N	Footer	Fixed BBh

CHANGE LOG

1. version 2.00 P

- a) Added this section.
- b) Changed 43h command
- c) Added command (47h)
- d) Removed definition of capture from command number (1Dh & 1Eh). now only alarm can be enabled or disabled. Capture option is moved to command number (47h)

2. Version 2.00 R

- a) Interlocking rule changed byte 17 from “RFU” to “No of doors”
- b) Interlocking rule added bytes 66 67 68 (fail safe / fail secure mechanism)

3. Version 2.00S

- a) Removed scope variables from all inputs in command Input Definitions (1Ch)
- b) Added scope variables in command Input Action (1Dh)
- c) Added canteen commands for (C0h to C4h)

4. Version 2.00T

- a) Added password reset command
- b) Added ble para command

5. Version 2.00U

- a) Modified ble parameter command

6. Version 2.00V

- a) Changed mifare config commands.

7. Version 2.00Y

- a) Changed 20h (display parameter command little bit) added card id display format in an rfu byte

8. Version 2.00Z

- a) Changed 2Dh command entirely..

9. Version 2.01

- a) Changed Display & brightness para command.
- b) Changed minor version of this document

10. Version 2.02

- a) Command 17h (Reader setup) > byte index [19] > replaced from RFU to Transaction enable / disable?

11. Version 2.03

- a) Command 4Ah (Finger Encryption added).

12. Version 2.04

- a) Command 4Bh (Extended User Information) added.
- b) Rule : Multiple Authentication (40h)Deprecated.
- c) Rule Added Group Authentication (40h) Added.

13. Version 2.05

- a) Command F9h (Enable / Disable Ethernet modules)added.

14. Version 2.06

- a) Added command 2Ch (volume para)

15. Version 2.07

- a) Added command 2Dh (User password/ pin reset/change)

16. Version 2.08

- a) Updated command 16h (Controller setup) . new parameter added, wiegand out enabled in controller mode

17. Version 2.09

- a) Corrected command 4Dh (user pin change), added byte for door id

18. Version 2.10

- a) Updated some finger print parameter values.

19. Version 2.11

- a) Added command EAh = Get pending event from service (previously EAh was designated for Push Canteen Data, but was not getting used, was duplicate of E7h, so removed header)

20. Version 2.12

- a) Added command Escort Rule setup (4Eh).
- b) Modified command User Extended Info (4Bh).
- c) Renamed Obsolete Bytes named Site Code (SC) in various commands to Extended ID. Commands modified are 03h,04h,07h,0Ch,0Dh,2Dh,32h,3Bh,3Dh,41h,4Dh.

21. Version 2.13

- a) Added command MQTT Publisher Rule setup (4Fh)

22. Version 2.14

- a) Modified Controller Setup command (16h / 17h) To indicate new type transaction frame
- b) Added a new command Temperate Sensor Para (50h).
- c) Added new push command New Type Transaction Frame (EBh).

23. Version 2.16

- a) Added command 51h

24. Version 2.17

- a) Added command D0h

25. Version 2.18

- a) Added command 52h, Mifare MAD Config / Paramters

26. Version 2.22

- a) Added command 53h, 54h related OSDP setup
- b) EAC reader setup command (55h) code id changed from ECh

27. Version 2.23

- a) Amended Matrix APB rule 38H

28. Version 2.25

- a) Added Command D1h (Server auth query & response).
- b) Added command 56h (Server auth rule setup).

29. Version 2.26

- a) Input to Output Trigger Command (1Dh). accommodate rule reset on GPI Trigger.

30. Version 2.27

- a) Added command 57h and 58h

31. Version 2.28

- a) Added Command, ECh (Push Log)
- b) Modified Command 48h

MEMORY_MANAGEMENT

1. Timezone Table

Byte No	Data	Value(d)
1	TZ 1 Start Hour (fixed read only)	0
2	TZ 1 Start Minute (fixed read only)	0
3	TZ 1 End Hour (fixed read only)	23
4	TZ 1 End Minute (fixed read only)	59
5 to ...		0 to 59
		0 to 59
		0 to 59
504	TZ 127 Start Hour	0 to 59
506	TZ 127 Start Minute	0 to 59
507	TZ 127 End Hour	0 to 59
508	TZ 127 End Hour	0 to 59
509	Check Sum_1st_Byte	0 to 255
510	Check Sum_2nd_Byte	0 to 255
511	Check Sum_3d_Byte	0 to 255
512	Check Sum_4th_Byte	0 to 255

2. TimeZone Xs Group.

Byte No	Data	Value(d)
1	Group1_1st Time Zone for Gate 1	1 to 127
2	Group1_2st Time Zone for Gate 1	1 to 127
3	Group1_3st Time Zone for Gate 1	1 to 127
4	Group1_4st Time Zone for Gate 1	1 to 127
5	Group1_Allow / Deny for Gate 1	00h : allow 01h : deny
6	Group1_1st Time Zone for Gate 2	
7	Group1_2st Time Zone for Gate 2	

8	Group1_3st Time Zone for Gate 2	
9	Group1_4st Time Zone for Gate 2	
10	Group1_Allow / Deny for Gate 2	
11	Group1_1st Time Zone for Gate 3	
12	Group1_2st Time Zone for Gate 3	
13	Group1_3st Time Zone for Gate 3	
14	Group1_4st Time Zone for Gate 3	
15	Group1_Allow / Deny for Gate 3	
16	Group1_1st Time Zone for Gate 4	
17	Group1_2st Time Zone for Gate 4	
18	Group1_3st Time Zone for Gate 4	
19	Group1_4st Time Zone for Gate 4	
20	Group1_Allow / Deny for Gate 4	
21	Group1_Check Sum_1st_Byte_	0 to 255
22	Group1_Check Sum_2nd_Byte	0 to 255
23	Group1_Check Sum_3d_Byte	0 to 255
24	Group1_Check Sum_4th_Byte	0 to 255
...		
...		
...		
24568	Group1_1st Time Zone for Gate 4	
24569	Group1_2st Time Zone for Gate 4	
24570	Group1_3st Time Zone for Gate 4	
24571	Group1_4st Time Zone for Gate 4	
24572	Group1_Allow / Deny for Gate 4	
24573	Group1_Check Sum_1st_Byte	
24574	Group1_Check Sum_2nd_Byte	
24575	Group1_Check Sum_3d_Byte	
24576	Group1_Check Sum_4th_Byte	

3.Week Of Table

Byte No	Data	Value
1	Group 1 Week 1 Days	b0XXXXXXX 0b0SFTWTMS
2	Group 1 Week 2 Days	
3	Group 1 Week 3 Days	
4	Group 1 Week 4 Days	
5 to ...	Group 1 Week 5 Days	
6 to ...		
157	Group 1 Week 1 Days	
158	Group 1 Week 1 Days	
159	Group 1 Week 1 Days	
160	Group 1 Week 1 Days	
161	Check Sum_1st_Byte	0 to 255
162	Check Sum_2nd_Byte	0 to 255
163	Check Sum_3d_Byte	0 to 255
164	Check Sum_4th_Byte	0 to 255

4.Holiday Xs Group Table

Byte No	Data	Value(d)
1	Group1_Date 1	1 to 31
2	Group1_Month 1	1 to 12
3	Group1_Year 1	00 to 99
4	Group1_Date 2	
5	Group1_Month 2	
6	Group1_Year 2	
7 to		

382	Group1_Date 128	
383	Group1_Month 128	
384	Group1_Year 128	
385	Check Sum_1st_Byte_Group1	0 to 255
386	Check Sum_2nd_Byte_Group1	0 to 255
387	Check Sum_3d_Byte_Group1	0 to 255
388	Check Sum_4th_Byte_Group1	0 to 255
389 to		
12028	Group32_Date 1	1 to 31
12029	Group32_Month 1	1 to 12
12030	Group32_Year 1	00 to 99
12031	Group32_Date 2	
12032	Group32_Month 2	
12033	Group32_Year 2	
12034 to		
12410	Group32_Date 128	
12411	Group32_Month 128	
12412	Group32_Year 128	
12413	Check Sum_1st_Byte_Group32	0 to 255
12414	Check Sum_2nd_Byte_Group32	0 to 255
12415	Check Sum_3d_Byte_Group32	0 to 255
12416	Check Sum_4th_Byte_Group32	0 to 255

5.ID Data base .

Byte No	Data	Value(d)
1	1st_User_UID LSB	0 to 255
2	1st_User_UID HSB	0 to 255
3	1st_User_UID USB	0 to 255
4	1st_User_UID MSB	0 to 255
5 to ...		
999997	2.5Lac_User_UID LSB	0 to 255
999998	2.5Lac_UID HSB	0 to 255
999999	2.5Lac_UID USB	0 to 255
1000000	2.5Lac_UID MSB	0 to 255

6.User Database

Byte No	Data	Value(d)
1	User_UID LSB	
2	User_UID HSB	
3	User_UID USB	
4	User_UID MSB	
5	SC_UID LSB	
6	SC_UID HSB	
7	SC_UID USB	
8	SC_UID MSB	
9	Time Zone Wise / Exclusive	
10	User Type	
11	Week Off group number LSB	
12	Week Off group number HSB	
13	Holiday Access group LSB	

14	Holiday Access group HSB	
15	Start Year	
16	Start Month	
17	Start Date	
18	Start Hour	
19	Start Minute	
20	End Year	
21	End Month	
22	End Date	
23	End Hour	
24	End Minute	
25	Pin LSB	
26	Pin HSB	
27	Pin USB	
28	Pin MSB	
29 to 60	User Name (32 Bytes)	
61 to 76	Employee ID (16 Bytes)	
77	Time Zone Group ID 1 LSB	(Item Xs Group)
78	Time Zone Group ID 1 HSB	
79	Time Zone Group ID 2 LSB	
80	Time Zone Group ID 2 HSB	
81	Time Zone Group ID 3 LSB	
82	Time Zone Group ID 3 HSB	
83	Time Zone Group ID 4 LSB	
84	Time Zone Group ID 4 HSB	
85	Time Zone Group ID 5 LSB	
86	Time Zone Group ID 5 HSB	
87	Time Zone Group ID 6 LSB	
88	Time Zone Group ID 6 HSB	
89	Time Zone Group ID 7 LSB	

90	Time Zone Group ID 7 HSB	
91	Time Zone Group ID 8 LSB	
92	Time Zone Group ID 8 HSB	
93 to 96	Escort ID (4 Bytes)	
97 to 100	Escort Site Code (4 Bytes)	

6. Fingerprint Data base :

Sector No.	Data
1	1st_User_UID Fingerprint_1 (384 Bytes)
2	1st_User_UID Fingerprint_2 (384 Bytes)
3	1st_User_UID Fingerprint_3 (384 Bytes)
4	1st_User_UID Fingerprint_4 (384 Bytes)
5	1st_User_UID Fingerprint_5 (384 Bytes)
6	1st_User_UID Fingerprint_6 (384 Bytes)
7	1st_User_UID Fingerprint_7 (384 Bytes)
8	1st_User_UID Fingerprint_8 (384 Bytes)
9	1st_User_UID Fingerprint_9 (384 Bytes)
10	1st_User_UID Fingerprint_10 (384 Bytes)
11	1st_User_Finger_Registration_Info (10 Bytes)
...	
...	
27,49,990	2.5st_User_UID Fingerprint_1 (384 Bytes)
27,49,991	2.5st_User_UID Fingerprint_2 (384 Bytes)
27,49,992	2.5st_User_UID Fingerprint_3 (384 Bytes)
27,49,993	2.5st_User_UID Fingerprint_4 (384 Bytes)
27,49,994	2.5st_User_UID Fingerprint_5 (384 Bytes)
27,49,995	2.5st_User_UID Fingerprint_6 (384 Bytes)
27,49,996	2.5st_User_UID Fingerprint_7 (384 Bytes)
27,49,997	2.5st_User_UID Fingerprint_8 (384 Bytes)
27,49,998	2.5st_User_UID Fingerprint_9 (384 Bytes)

27,49,999	2.5st_User_UID Fingerprint_10 (384 Bytes)
27,50,000	2.5st_User_Finger_Registration_Info (10 Bytes)

7. Item Database :

Byte No	Data	Value(d)
1	Item Number LSB	
2	Item Number MSB	
3	Active / Inactive	
4	Item Price (LSB)	
5	Item Price (HSB)	
6	Item Price (USB)	
7	Item Price (MSB)	
8	Item Name Byte1	
...	...	
23	Item Name Byte16	
24	Week Day Group	
25	Canteen Code	
26	Vendor Code	

8. Item Group :

Byte No	Data	Value(d)
1	Location ID	
2	Item Group Number	
3	Prepaid / PostPaid	
4	Item Number LSB	
5	Item Number MSB	
6	Timezone1 Number	

7	Timezone2 Number	
8	Timezone3 Number	
9	Timezone4 Number	
10	Item Restriction Type	
11	Item Restriction Quantity	
12	Subsidize price LSB	
13	Subsidize price MSB	
14	Subsidize price USB	
15	Subsidize price HSB	
16	Subsidize Restriction Type	
17	Subsidize Restriction Quantity	

9. Vendor Database:

Byte No	Data	Value(d)
1...	Vendor1 Name Byte1	
...32	Vendor1 Name Byte32	
33...	Vendor2 Name Byte1	
...64	Vendor2 Name Byte32	
...		
...		
...		
4064...	Vendor128 Name Byte1	
...4096	Vendor128 Name Byte32	